CHAPTER SIX

Sensors

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Sensors on-board a spacecraft are the source for any information, measurement, or data used by the guidance, navigation, and control (GNC) system to accomplish its task. In analogy with our human experience, they are the sensory organs (e.g., eyes, ears, nose, etc.) sensing the external environment to acquire information needed for the GNC operations. A proper GNC design shall select and arrange the sensors on the spacecraft system in a way that all the desired quantities are properly measured. In doing so, the GNC designer shall know, understand, and correctly model the sensors to support its design process. This chapter is dedicated to briefly present all the major sensor typologies for spacecraft GNC applications, to discuss their operating principles, to highlight their peculiarities, to categorize the available components according to their performances, and to discuss the most relevant modeling techniques and rules.

The content of this chapter is structured as follows:

- *Sensor modeling for GNC.* In this section, the sensor modeling techniques are presented, discussing the main implementation principles of the most common numerical models to include the sensor behavior in the GNC design, analysis, verification, and validation phases. The main error sources and their modeling are introduced, together with the most typical sensor faults. Moreover, this section contains some elements of metrology, and it explains some basic principles of statistics and random variables representation.
- *Orbit sensors.* This section presents the Global Navigation Satellite System (GNSS) and the ground-based sensors for orbit GNC. Their main operating principles and the available performance are discussed. A simple GNSS sensor model implementation in MATLAB/Simulink is included in this section.

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- *Attitude sensors*. This section introduces the main typologies of sensors for attitude GNC. For each of them, it presents the main characteristics and the operational principles. The available performance of the different attitude sensors is compared, and an example Sun sensor model implemented in MATLAB/Simulink is also described.
- *Inertial sensors*. Accelerometers and gyroscopes are presented in this section. Their peculiar error sources are listed and presented, together with the available performance according to the quality grade of the sensor. A brief introduction to the Allan variance (AVAR) representation of their random errors is included. The chapter is concluded with a simple example of a MATLAB/Simulink implementation of a gyroscope model.
- *Electro-optical sensors*. This section introduces the sensors with a vision-based sensing capacity, such as cameras and Light Detection and Ranging (LIDAR) systems. The two typologies, respectively, represent the class of passive and active electro-optical sensors. Their applicability range and their design principles are briefly discussed.
- *Altimeters*. In this section, the concept of altimetry is introduced and the most relevant typologies of altimeters for spacecraft applications are discussed. A mathematical altimeter model is also presented at the end of this section.

Sensor modeling for GNC

Correct sensor modeling is of utmost importance for GNC applications and design. Indeed, algorithms development and testing shall rely on proper sensor models to have a truthful representation of the available measurements. Moreover, some estimation filters are based on measurement models that are capable to include the terms to be estimated. Thus, we need to understand what the model of a sensor is, how it is built, and how it is used along the GNC design and verification processes.

A sensor is an electronical device exploiting a physical principle to measure a certain physical quantity, such as the angular velocity, the position of the spacecraft, or the position of a celestial body, among the others. The external reality is sensed by the sensor's detector, elaborated by the sensor processing units, and communicated to the GNC system via electrical signals, which can be analogic or digital, even if modern spacecraft applications are almost uniquely based on digital components. Along this entire process, the measured physical quantity is inevitably distorted and modified with

respect to the reality, and a good sensor model shall be able to represent these deviations, together with all the effects due to the sensing activities. We are going to mathematically refer to this as:

$$\breve{\mathbf{x}}(t) = \mathbf{x}(t) + \boldsymbol{\varepsilon}(t), \tag{6.1}$$

where $\breve{\mathbf{x}}(t)$ is the measured quantity, $\mathbf{x}(t)$ is the real-world true quantity, and $\boldsymbol{\varepsilon}(t)$ is the summation of all the sensing process errors and effects.

A sensor model is an abstraction of the actual sensing process converting the physical quantity into the electrical output, and it is implemented for a precise purpose with specific objectives. In fact, there exist different types of sensor models: operational, behavioral, functional, and performance models are the most common.

Operational sensor models are used to describe and represent how the sensor is operated and interfaced with the rest of the GNC system, in terms of commands, operational modes, and interfaces with other components. Behavioral models reproduce the global behavior of the sensor, including transitions between modes, response to inputs, and evolution of the outputs. However, behavioral models do not model the internal processes of the sensor, but they describe an averaged transient or steady-state behavior. Functional sensor models are established to serve as input for detailed analyses and accurate performance simulations. They outline all the internal functions executed along the sensor processing, being representative of the sensor actual temporal performances for realistic output profiles. The complete equipment acquisition chain is typically modeled, from signal transduction and conditioning to digitalization and formatting. Functional models can be representative of real input and output data formats, but they are not required to include different operational modes. Consequently, more functional models may be necessary to simulate the system in its entire operational domain, with all its different modes. Performance sensor models represent the way in which the sensor operates and interacts with the GNC system, in terms of performances, consumed resources, and physical limitations. Thus, they are not focused on the operations, functions, and behaviors of the sensor, but on the implications the performance of the sensor has on the surrounding system. They are used to verify how the GNC system works, highlighting potential bottlenecks of the design, given the performance of the selected components. Performance models use equivalent generic mathematical models, and they are commonly modeling only the principal and the most influent characteristics of the sensors.

Functional and performance sensor models are the most used for GNC design, analysis, simulation, and verification. In fact, they respectively allow to assess the detailed functionalities and the overall performance of any element of the system. Moreover, specific analyses and verification procedures may require a combination of the features of the aforementioned sensor models. For example, an operational-functional sensor model can be used to verify with great accuracy a GNC subsystem in diverse operative modes. In addition, there are very specific and application-oriented sensor models for telemetry and telecommand (TMTC),i.e., models to generate sensor telemetry packets or react to telecommands, or for failure mode, effects, and criticality analysis **(**FMECA) analyses, i.e., models to simulate faults or effects due to failures.

Sensor model's interfaces need to be defined according to the specific type of model in use and considering the desired application. The interfaces of the sensor model determine the input/output of the system simulator, which is used for GNC design and verification, and their fidelity level shall be selected accordingly. In fact, the sensor model's interfaces may be formed by representative engineering data or by the raw sensor data. The former are numeric values, whose data format is not relevant for the specific application, and they are directly used by the algorithms in the mathematical formulas. The latter are representative of the raw values, as communicated by the real sensor hardware with a specific data format. Thus, they need to be processed and converted in the desired format for the following operations. For this reason, raw sensor data interfaces are typically used when the complete GNC verification is of interest, with particular focus on the verification of GNC software components. Alternatively, engineering data are handy when the GNC design is on-going, and the analyses are focused more on the GNC algorithms performance than on the final development and implementation.

Sensor models are exploited both in accelerated and in real-time simulators. Accelerated simulations typically support the early development and verification stages, while the real-time models are integrated in dedicated testing benches to verify the latest hardware and software developments, as will be extensively discussed in Chapter 12 — GNC Verification and Validation. It is convenient to implement a model having in mind both applications, since it is very common to port the model from the accelerated simulation environment (e.g., MATLAB/Simulink) to the real-time one. In these regards, the requirements for the real-time model (e.g., compatibility with auto-coding techniques, compatibility with real-time

target and execution) should be considered since the beginning of the development in the accelerated simulation environment. The two typologies of simulated models shall be validated together to confirm that they produce the same results, and they have the same temporal evolution. In particular, the execution order of the accelerated simulation shall be carefully checked to ensure coherence with the real-time application. This last point may be facilitated if the sensor model is developed using atomic unit blocks [1] in accordance with the real sensor processing steps.

Like any other part of a GNC system, the sensor models shall be validated, meaning that they shall be truly representative of the real sensor component. The validation campaign shall be focused both on the short-time and on the long-time evolution of the sensing process errors and effects, noting that the long-time ones have a deeper impact on the GNC performance. Ideally, the validation should also address the variation of the sensor behavior with the external environment (e.g., temperature, pressure, acceleration state). A sensor model is typically developed exploiting the data available in the datasheets and in the documentation provided by the producer. However, in the later stages of the GNC development and verification, the model parameters must be updated in accordance with real measured sensor data. Note that the data obtained from the exact flight model of the sensor installed on the spacecraft should be used for a thorough and comprehensive validation campaign.

The primary use of the sensor models is the support of GNC design and verification emulating the presence of the sensors to be installed on-board. However, the sensor models are also directly used in the GNC blocks. For instance, guidance functions to calibrate or initialize sensors, navigation filters measurement models, control methods to reduce effects disturbing a particular sensor rely on a correct modeling of the sensor measurements. Hence, as will be discussed in Part 2 – Spacecraft GNC, it is common to use dedicated sensor models or specific sensor variables directly inside the GNC functions.

Elements of metrology

Sensors convert the physical quantity to measure into an electrical signal following a predefined sequence of transduction and processing steps, which

is denoted as sensor acquisition chain. It depends on the actual implementation of the specific sensor, but it is generally composed by:

- Transduction.
- Conditioning.
- Analog-to-digital (AD) conversion.
- Calibration and transmission.

Transduction is the process of converting the physical property into the corresponding electrical signal. Conditioning is needed to process the transduced sensor signals into a form that can be converted to digital values, and it is typically composed by signal filtering, amplification and shaping, and linearization. AD conversion allows the digitalization of the conditioned electrical signals by converting the analog input voltage or current to a digital number representing the magnitude of the voltage or current. Then, the digital signal can be calibrated to convert back the electrical data into the equivalent physical quantity that has been measured. Finally, the calibrated digital signals are encoded and formatted to reduce transmission errors and to guarantee compatibility with the communication protocol of the sensor. Fig. 6.1 summarizes a generic sensor acquisition chain. Note that sensors for space applications commonly transmit their data by exploiting typical communication interfaces, such as Universal [Synchronous and] Asynchronous Receiver-Transmitter (U[S]ART), Serial Peripheral Interface (SPI), Inter-Integrated Circuit (I2C), Controller Area Network (CAN), etc.

Sensors can measure absolute quantities or differential ones. In the first case, which is the most common for spacecraft applications, the sensor is designed to seek the "true" value of the physical quantity to measure. While, in the case of differential sensors, the measurement is performed to seek a deviation with respect to a "reference" value. This is done to minimize systematic errors, which are affecting both the reference and the differential measurements. Note that in absolute sensors, the measurement error is the deviation between the obtained measurement result and the true value, while differential sensors have the errors affecting the difference between

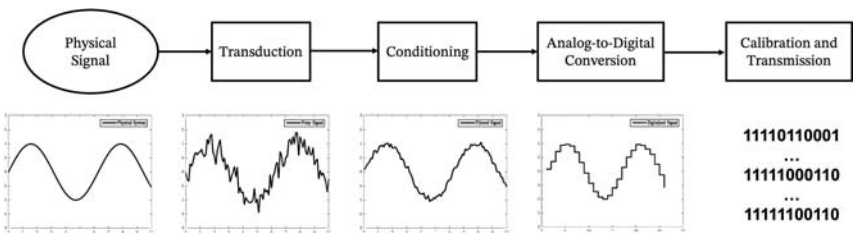


Figure 6.1 Sensor acquisition chain.

the reference and actual measurements. As a result, good differential measurements can be obtained even with a system of poor quality. Anyhow, the following discussion will be focused on absolute sensors.

Measurement errors, $\boldsymbol{\varepsilon}(t)$, are impossible to be exactly known, but they can be assessed under a statistical perspective. Thus, we shall introduce the concept of measurement uncertainty, which gives us information about the amount of measurement error that is likely to occur. The measurement uncertainty is defined as a "nonnegative parameter characterizing the dispersion of the quantity values being attributed to the measured physical quantity" [2]. In fact, the measurement result is a random variable, which is described by a probability density function that is typically bell-shaped (i.e., normal statistical distribution). An ensemble of more measurement results is centered at a certain statistical average value and dispersed according to a statistical standard deviation (or variance). These two quantities are associated to two concepts about sensors: accuracy and precision.

The output of the sensor is the measure, $\breve{\mathbf{x}}(t)$, which is, by definition, different from the real measured quantity. The closer the measure is to the real world, the more accurate the sensor is. Indeed, the accuracy of a sensor is how close the measurement is with respect to truth (i.e., the distance of the statistical average value with respect to the true value). The precision of a sensor is the reproducibility with which a measurement is made (i.e., the dispersion of the normal distribution, related to the statistical standard deviation). We shall not confuse the definitions of these two concepts, visually represented in Fig. 6.2. Note that for each realization of the measurement, there is a single resulting error value, which relates each measure to the actual value according to Eq. (6.1). Hence, the statistical quantities behind accuracy and precision combine themselves identifying the interval containing the measurement result values.

In general, the main concepts characterizing the sensors are:

- *Accuracy*. The mean difference that will exist between the actual value and the indicated value at the output of the sensor.
- *Precision*. The degree of reproducibility of a measurement, which is associated to closeness between measured quantity values obtained by replicate measurements on the same quantity.
- *Sensitivity*. The minimum input of the physical parameter that will create a detectable output change, which is associated to the slope of the input-to-output characteristic curve (i.e., output change for a given change in the input).

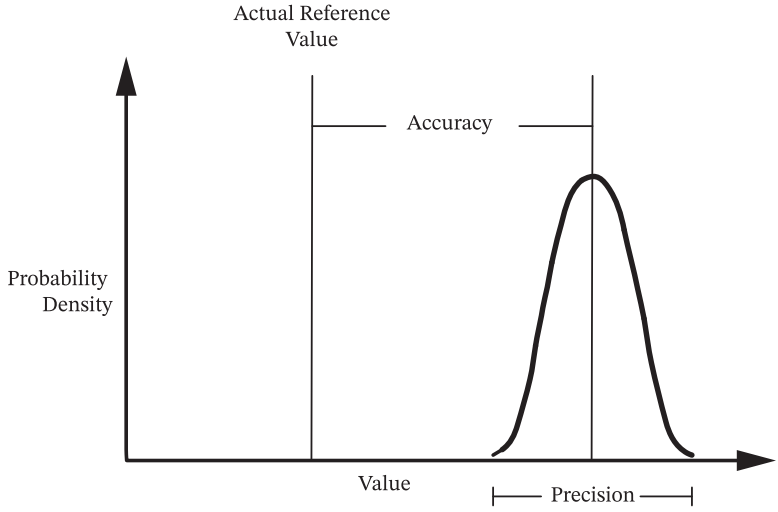


Figure 6.2 Sensor accuracy and precision.

- *Range*. The maximum and minimum values of the applied parameter that will be measured.
- *Resolution*. The smallest detectable incremental change of input parameter that will be detected in the output signal.
- *Offset* (*bias*). The output that will exist when it should be zero.
- *Linearity*. The deviation the actual measured curve of a sensor will have from the ideal input-to-output linear characteristic curve.
- *Hysteresis*. The difference the actual measured curve of a sensor will have as a function of the input variation direction.
- *Response time*. The time required for a sensor output to change from its previous state to a final settled value within a tolerance band of the correct new value.
- *Dynamic linearity*. The ability of a sensor to follow rapid changes in the input physical parameter.

All the previous concepts characterize the behavior of the sensor, and they shall be included in a proper sensor model.

Inferring from the previous discussion, we understand how precision is a description of random errors (e.g., noise, instabilities) and accuracy is a description of systematic errors (e.g., offset, imperfect calibrations). Before entering the details of both error typologies, we shall focus on the fundamentals of probability and stochastic processes, and on the systematic errors induced by a wrong calibration.

Probability and stochastic processes

Random errors require a statistical interpretation to be understood, and the fundamental concept behind this is the probability. By definition, the probability of an event A to occur is defined by how likely this event is to happen and it can be generally defined as:

$$P(A) = \frac{\text{number of times } A \text{ occurs}}{\text{total number of occurrences}}.$$

However, the probability of a given event A to occur can be also linked to an event B that happened before. In this case, we talk about conditional probability of A given B and we can define it as:

$$P(A|B) = \frac{P(A,B)}{P(B)}$$

where $P(A,B)$ is the joint probability of A and B and it describes the probability that both A and B occur.

Random variables

Another very important concept to introduce when dealing with the statistical interpretation of random errors is the one of random variable. A random variable is a mathematical description of the outcome of random events. In a broader sense, it can be seen as a variable mapping a set of random events to a set of real values. The most interesting thing about random variables is that they have associated properties very useful in the domain of probability and statistics. In particular, one of the most important quantities associated to a random variable is the probability distribution function (PDF), defined as:

$$F_X(x) = P(X \leq x)$$

with, in this case, X being a generic random variable and x a nonrandom variable or constant. Similarly, the probability density function (pdf) can be defined as:

$$f_X(x) = \frac{dF_X(x)}{dx}.$$

Also for random variables, conditional quantities can be defined, given that a certain event A has occurred:

$$F_X(x|A) = P(X \leq x|A),$$

$$f_X(x|A) = \frac{dF_X(x|A)}{dx}.$$

In order to quantify the average value of a given random variable over a large number of experiments, it is necessary to introduce the concept of expected value. The expected value of a random variable X, or mean of X, is defined as:

$$E(X) = \mu = \frac{1}{N}\sum_{i=1}^{m} A_i n_i = \int_{-\infty}^{\infty} x f_X(x) dx$$

where N is the number of total experiments, m is the total number of possible different outcomes, and A_i is a given event occurring n_i times. Please recall that $f_X(x)$ is the pdf of X. Using the definition of expected value, it is possible to also define the concept of variance. The variance of a random variable is a measure of how a given random variable will vary from its expected value or mean. The variance of a given random variable X is defined as:

$$\sigma_X^2 = E\left[(X-\mu)^2\right] = \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x) dx$$

where σ_X is also known as standard deviation of a random variable.

In general, a random variable X can be defined as $X \sim (\mu, \sigma^2)$, having mean μ and variance σ^2.

Different types of random variables and PDFs exist and have been categorized in statistical literature [3]. Here, we will recall the most important ones for spacecraft GNC applications.

Uniform random variables

A uniform random variable has, by definition, a pdf that is constant between two limits a, b. In practical terms, the random variable X has an equal probability of assuming any value in the interval defined by the pdf limits and zero probability outside the interval. In a mathematical way, it is defined as:

$$f_X(x) = \begin{cases} \dfrac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

A graphical representation of a uniform distribution is shown in Fig. 6.3.

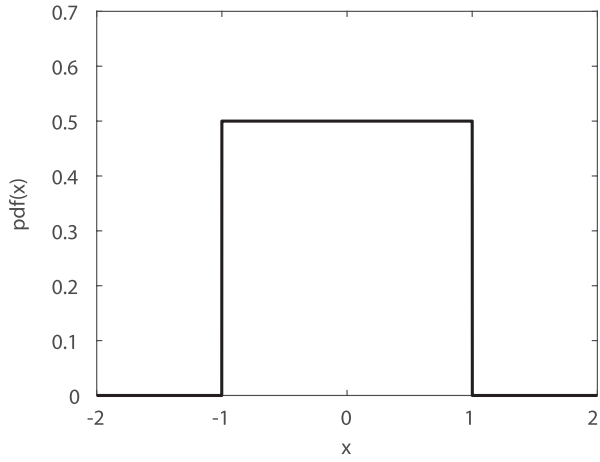


Figure 6.3 Uniform distribution.

Gaussian random variables

A random variable is defined Gaussian or normal if its pdf is expressed as:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[\frac{-(x-\mu)^2}{2\sigma^2}\right]}$$

The graphical representation of a zero-mean Gaussian normal distribution with unitary variance is reported in Fig. 6.4.

An important consideration about Gaussian random variables is that, with this distribution, the variable will have 68% of probability to assume a value:

$$X = \mu \pm 1\sigma$$

In other words, there will be 68% odds for the variable to lie within one standard deviation from the mean, 95% of odds to lie within two standard deviations from the mean, etc. It is a normal practice to define the error as being specified at 1σ, or, more conservatively, at 3σ, or whichever level. Recalling Fig. 6.2, it can also be interpreted as the likelihood for the error to assume that value. As an applicative example, if the designer models the random error as a constant noise with its 3σ value, it means that it will overestimate the actual error 99.7% of the times.

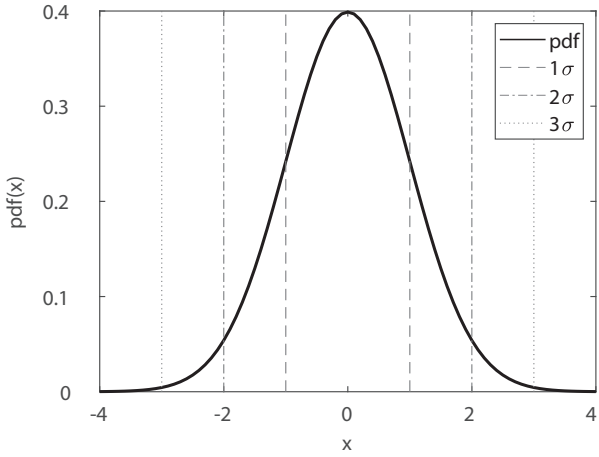


Figure 6.4 Gaussian or normal distribution.

Stochastic processes

The variation of a random variable in time can be represented through a stochastic process. A stochastic process $X(t)$ describes the variation in time of a random variable X. As previously done for a random variable, PDF, pdf and expected value of a stochastic process can be defined. In particular, the PDF takes the form of:

$$F_X(x,t) = P(X(t) \leq x)$$

and for the pdf:

$$f_X(x,t) = \frac{dF_X(x,t)}{dx}.$$

Similarly:

$$E(X(t)) = \mu = \int_{-\infty}^{\infty} x f_X(x,t)\,dx$$

$$\sigma_X^2 = E\left[(X-\mu)(X-\mu)^T\right] = \int_{-\infty}^{\infty} (x-\mu)(x-\mu)^T f_X(x,t)\,dx$$

While talking about stochastic processes, if a random variable at a given time t_1 is not dependent on its value at another instant of time t_2, we call this stochastic process as white noise. Otherwise, we talk about colored noise,

with different types of colors depending on the stochastic process power spectrum. These concepts will be deepened in the following of this chapter, when discussing about sensor noise and random errors. Moreover, the reader is invited to extend its knowledge on statistical methods in literature references [3,4] and in the appendix Chapter 16 — Mathematical and Geometrical Rules.

Sensor calibration

Sensor calibration is crucial for GNC applications, since a good calibration campaign can minimize most of the systematic errors, improving the system's accuracy up to the best nominal accuracy of the selected sensors. It should be remarked that the usage of accurate sensors without a proper calibration is a huge mistake under the engineering perspective and a waste of resources under the programmatic perspective. Sensor calibration is the comparison of measurement values delivered by the sensor with those of a calibration reference standard. Strictly speaking the calibration definition does not include the adjustments to improve the measurement accuracy, but it is commonly implied that the calibrating adjustments are performed along the calibration procedures. Spacecraft sensors are calibrated to remove statistical biases introduced by mounting errors, axis misalignments, temperature dependencies, geometrical distortions, environmental influences, scale factors, nonlinearities, and other potential sources of systematic errors. A complete calibration can be very complex, and a trade-off shall be performed between achievable accuracy and calibration effort. Note that a perfect calibration is practically not achievable because of the intrinsic finite accuracy of the calibration reference standard and the finite precision of the sensor. Moreover, it is suggested to have a calibration effort that is coherent and balanced with the precision of the sensor, the overall accuracy of the other sensors on the spacecraft, and the required accuracy of the GNC system.

The calibration shall be performed on-ground during the GNC development, especially for those errors related with the assembly and integration phases (e.g., misalignments, distortions). However, any calibration is not perpetual, and successive recalibrations shall be performed at prescribed intervals of time. If this is very common and standardized for Earth applications, the same is not true for spacecraft due to the physical inaccessibility of the platform. So, we should integrate calibration standards on-board or use the known space environment — or the spacecraft dynamics — as

calibrating references. In any case, on-board sensor calibration is suggested for those applications looking for a high accuracy. It can be performed at discrete intervals, or it can be continuously executed. Navigation and estimation functions are capable to quantify biases on-board to recalibrate the sensors, as will be discussed in Chapter 9 — Navigation and in the on-board sensor calibration of Chapter 14 — Applicative GNC Cases and Examples.

Errors modeling

Sensor models for GNC applications are based on mathematical models of the processes along the sensor acquisition chain. They take as input a simulated physical signal, and they are set to output the sensor measurements. The format and the characteristics of the output depend on the typology of sensor model, which also influences the way in which the internal processes are modeled. As said, the most complete functional models include almost every step of the acquisition chain, as shown for a generic functional sensor model in Fig. 6.5. Note that other typologies of sensor models may be composed by equivalent subsets of the blocks used for a functional model.

With reference to Fig. 6.5, the most relevant systematic and random error sources for spacecraft sensors are:

- Bias or zero offset errors.
- Scale factor errors.
- Noise or random fluctuations errors.
- Quantization or resolution errors.
- Misalignment and nonorthogonality errors.

Moreover, other sensor processing phenomena exist, and even if they cannot be defined as errors, they shall be included in proper sensor models:

- Time delays and latencies.
- Maximum range or output saturation.
- Output temporal discretization.

The sensor acquisition chain is also influenced by the environment (e.g., temperature, acceleration, radiations, etc.). The environmental influences

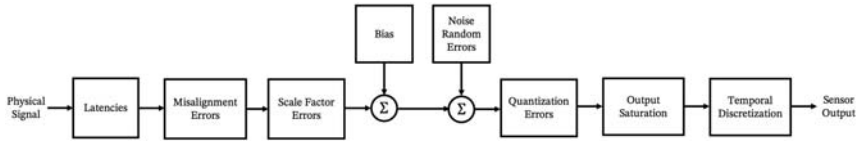


Figure 6.5 Generic functional sensor model example.

are not immediately evident from the previous discussion, since the changes in the external environmental conditions affect the sensor output in an indirect way. That is, the previously listed error terms are dependent from the external environment. For example, the bias and the scale factor errors may be dependent from the accelerations, the noise may vary as a function of the temperature, or other similar dependencies. In general, the cross talk between potentially interacting physical quantities (e.g., acceleration, magnetic field, angular velocity, temperature, etc.) shall be carefully investigated and accounted during the implementation of the sensor models.

Sensor models may be also influenced by additional aspects, such as aging, contamination from pollutant, light source reflections, and occultations. Thus, also these effects shall be modeled and considered, if relevant for the specific application.

Finally, note that sensors whose processing is dependent from the time (e.g., GNSS sensors) may be subject to indirect errors because of clock inaccuracies or time deviations with respect to the universal timing.

The most relevant error terms are briefly described in the followings, whose modeling numerical parameters are commonly found in the datasheet of the sensor.

Bias

Measurement bias is a systematic error that set an offset in the output signal with respect to the actual physical value. Typically, bias randomly varies at each start-up of the sensor, and it is influenced by the external environment. It is assumed to be constant or slowly varying in time, and its temporal fluctuations are categorized within the random error sources. It is one of the error sources most prone to be minimized, thanks to calibration.

Its mathematical model is:

$$\breve{\mathbf{u}} = \mathbf{u} + \mathbf{b},$$

where $\mathbf{b}$ is randomly selected for each simulation around the nominal value stated in the sensor's datasheet.

Scale factor errors

Scale factor errors are a class of systematic errors that are proportional to the magnitude of the input signal. Scale factor errors may be of linear, nonlinear, or asymmetric type. Linear scale factors (or just scale factors) introduce a nonunitary slope coefficient between output and input. Nonlinear scale

factors are proportional to the square, or higher order terms, of the input, and they let the output curve deviate from a straight line. Asymmetry scale factors are proportional to the absolute value of the input, and they create an angle in the output curve close to the zero value. Scale factors may be influenced by the external environment, especially temperature, and should be modeled in the entire operative range of the sensor. Calibration techniques are effective in reducing scale factor errors.

Scale factor models can be mathematically described as:

$$\breve{\mathbf{u}} = \mathbf{u} + \mathbf{SF}\mathbf{u} + \mathbf{SF}_{\boldsymbol{N}}\mathbf{u}^2 + \mathbf{SF}_{\boldsymbol{A}}|\mathbf{u}|,$$

where $\mathbf{SF}$, $\mathbf{SF}_{\boldsymbol{N}}$, and$\mathbf{SF}_{\boldsymbol{A}}$, are, respectively, the linear scale factor, nonlinear scale factor, and asymmetry scale factor coefficients. These coefficients are usually expressed in parts per million, PPM. Note that 1 PPM corresponds to $10^{-6} = 0.0001\%$ of the input value.

Noise and random errors

Random errors are nonsystematic errors that influence any sensor, since they are intrinsically related to the stochastic nature existing in real-world applications. Because of their fundamental property, they are impossible to be fully compensated and a residual stochastic measurement error is present in any measure. In fact, the only way to reduce the effects of random errors is to average the measure from a large set of measurements. This is not always possible, since this procedure introduces a delay in the measurement processing that may be not acceptable for the specific GNC application. We will refer to noise alternatively to random error or stochastic process. In general, noise is used as a generic term to indicate any undesired errors generated by a stochastic process.

Random errors shall be described in terms of their statistical parameters, since a deterministic description is ontologically not possible. They can be of various typologies, but they are always characterized by an unpredictable and not replicable time evolution. Thus, they are associated to a standard deviation, σ, or variance, σ^2, but they commonly have null average because constant errors are taken into account together with bias errors. The probability distribution of random measurement errors is typically well approximated by a Gaussian normal distribution since, according to the central limit theorem, they are the result of the sum of many independent random variables. This can also be seen in Fig. 6.6, where a noise signal with $\sigma = 3$ is sampled for 100 s at 100 Hz. From the figure it can be seen how the noise signal samples

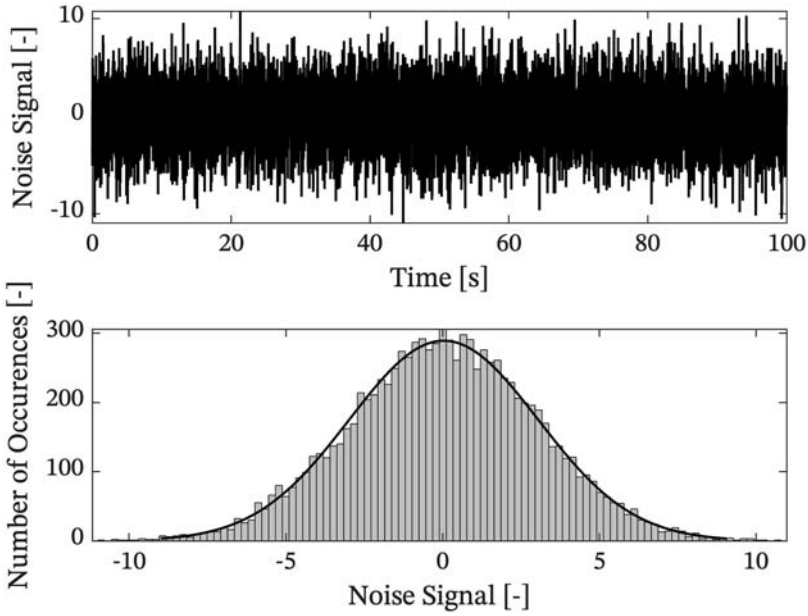


Figure 6.6 Noise signal with $\sigma = 3$.

are well fitted by a normal distribution with the same standard deviation. As a result, $\sim$68% of the samples are within the σ interval (i.e., $[-3,\ 3]$), while $\sim$99.7% of them are contained in the 3 σ interval (i.e., $[-9,\ +9]$). Indeed, almost all the samples have an absolute value lower than 10.

Random errors are also associated with different frequencies of the stochastic process behind them, and their resulting output can be generally identified by analyzing their power spectrum. A signal whose spectrum has equal power (i.e., intensity) within any equal interval of frequencies is defined as white noise; its power spectral density is constant. White noise is the most common noise source in sensor measurements, since it is usually introduced by the electronic sensor processing. However, there exist other random errors with a frequency spectrum that is not constant. They are defined as colored noises, and according to the shape of their power spectral density, they are associated with a color. For instance, pink noise has a power spectrum which decreases linearly in the logarithmic scale, while blue noise increases. Other colors may be defined, but they cannot be generalized. The most common noises that are relevant in sensor modeling are:

- White noise.
- Pink noise, or flicker noise, power spectral density decreases proportionally to $1/f$, where f is the frequency.
- Brown noise, or Brownian noise, or red noise or also identified as random walk noise power spectral density decreases proportionally to $1/f^2$, and it is the noise associated with the random walk errors affecting many digital sensors. Brownian noise can be generated with temporal integration of white noise.

An example of these noise sources with $\sigma = 10$ is reported in Fig. 6.7, in terms of their signal sampled for 1000 s at 100 Hz, and their power spectral density.

Noise modeling shall exploit random number generators from a normal distribution, defining the standard deviation and a parameter, usually denoted as seed, to initialize the random generator. Note that seed values shall be different between different components in the model, and there should be the possibility to change their value at each simulation to avoid simulating always an identical scenario. The plain normal random generator can model a white noise source. Filtering the random number generator's output can be used to simulate colored noise. For example, a low-pass filter with a power roll-off of 3 dB per octave (i.e., the signal power reduces by a factor of 2 every time the frequency doubles) can be applied to a random number generator to model a pink noise. Similarly, a low-pass filter with a power roll-off of 6 dB per octave can be used to model a brown noise.

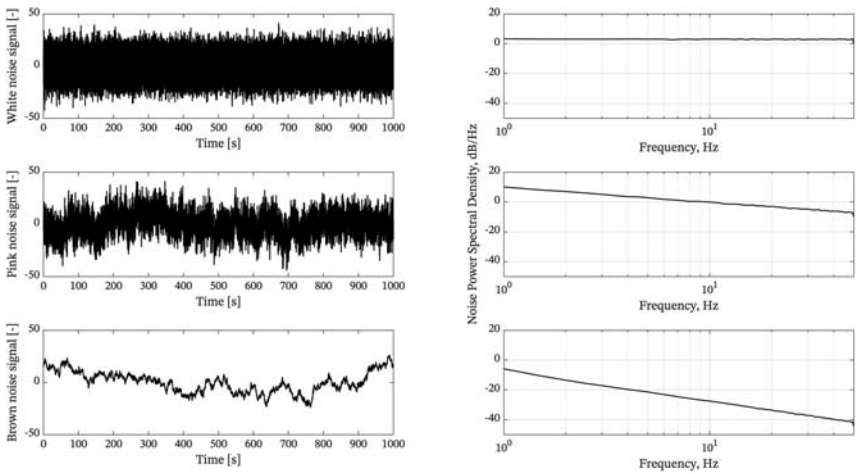


Figure 6.7 White, pink, and brown noise and their power spectral densities.

The low-pass filter should be set with a cut-off frequency at −3 dB as low as possible in order to be applied on the entire frequency spectrum. The cut-off frequency may be slightly varied to precisely tune the noise frequency content. Note that brown noise can be also modeled with a time integrator placed at the output of the random number generator. This is commonly done to model bias instabilities and fluctuations in scale factor errors. Example noise models are shown in Fig. 6.8.

Noise amplitude is tuned by varying the standard deviation in the random number generator. However, sensor datasheets do not frequently report this value, and the fine tuning of the sensor model shall be supported by sensor characterization testing activities. This is especially true for colored noise terms. Noise density is more often reported on the datasheets. In fact, the actual noise amplitude frequently depends on the sampling time of the sensor. Let's assume the noise amplitude, A, expressed in the units of the measured quantity, u, then the standard deviation of the normal random numbers shall be:

$$\sigma = A \text{ u}.$$

The noise amplitude density, D, is measured in units over the square root of the frequency, f (i.e., $\text{u}/\sqrt{\text{Hz}}$), and it can be related to the amplitude as:

$$A \text{ u} = D\sqrt{\frac{f}{2}} = \frac{D}{\sqrt{2T_s}},$$

where T_s is the sampling time of the sensor.

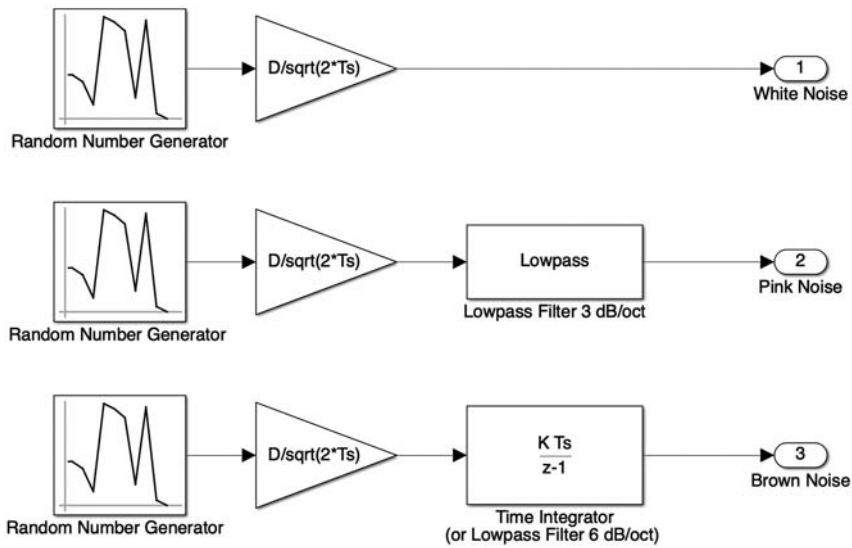


Figure 6.8 Noise models.

Random errors with uniform distribution

The Gaussian normal distribution is not the one for all PDF, since stochastic processes may be distributed according to different PDFs. As said, Gaussian distribution is important for engineering applications because of the central limit theorem, which states that, when measuring real-world data, the different measured values tend to produce a final distribution that is approximately normal. Generally, the more a measurement is like the sum of independent variables with equal influence on the result, the more normality it exhibits. Indeed, we can often regard a single measured real-world data value as the weighted average of many small independent effects.

However, there are situations in which this condition does not hold, and the random errors shall not be modeled by exploiting the normal distribution. Binomial distribution, Poisson distribution, uniform distribution, exponential distribution, and the Weibull distribution are examples of alternative PDFs. The former two are often used to represent the statistical distribution of discrete data, while the latter applies to continuous valued data [4]. They are often used in various specialized applications, but the most relevant among them for spacecraft GNC is the uniform distribution.

The uniform distribution was introduced before in the random variables section, and it is used to represent those data in which every value within a certain interval has the same probability to occur. For instance, the position of the center of mass during the design process of a spacecraft will be typically assumed to fall inside a box with uniform distribution, since any point within the acceptability range has the same probability to occur. Similarly, the uncertainties on the values of other geometrical or inertia properties of the spacecraft are commonly represented with a uniform distribution. In the same way, if the GNC design considers an initial condition for the spacecraft attitude dynamics after the deployment from the launcher, these are typically uniform in terms of the attitude states. In fact, all the rotations of the body frame with respect to the inertial one are possible.

In general, for any random variable, the GNC designer shall select the proper PDF. As a rule of thumb, measured real-world data are normally distributed, while the uncertainties in the system parameters or the unknown initial conditions are uniformly distributed within the acceptability ranges.

Quantization errors

Quantization error is a systematic error resulting from the difference between the continuous input value and its quantized output, and it is like

round-off and truncation errors. This error is intrinsically associated with the AD conversion that maps the input values from a continuous set to the output values in a countable set, often with a finite number of elements. The quantization error is linked to the resolution of the sensor. Namely, a high-resolution sensor has a small quantization error. Indeed, the maximum quantization error is smaller than the resolution interval of the output, which is associated to the least significant bit representing the smallest variation that can be represented digitally:

$$LSB = \frac{FSR}{2^{N_{BIT}}}$$

where FSR is the full-scale range of the sensor, and N_{BIT} is the number of bits (i.e., the resolution) used in the AD converter to represent the sensor's output. Quantization errors are typically not corrected, and the discrete values of the output are directly elaborated by the GNC system, which is designed to operate on digital values.

Fig. 6.9 shows a convenient model block to simulate quantization errors.

Misalignment and nonorthogonality errors

Misalignment and nonorthogonality errors are systematic errors due to mounting errors of the sensor, internal misalignment in the sensor's axes, or nonorthogonality between the different axes of an N-dimensional sensor. These errors can be easily minimized via ground calibration during the testing phase on the integrated spacecraft. However, residual and unknown misalignment and nonorthogonality are anyway present in the sensor measurements. Misalignment makes a measure to be not correctly associated to a nominal reference axis. Nonorthogonality determines a cross-axes sensitivity, since a physical quantity along an axis has nonzero spurious measured components also on the other axes of the sensor. Moreover, the nonorthogonality errors typically account also for the cross-axes electrical interference between the different axes of the sensors.

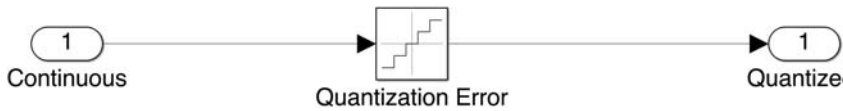


Figure 6.9 Quantization error model.

Misalignment and nonorthogonality errors are mathematically represented by multiplying the physical signal by rotation, **R**, and nonorthogonal, **O**, matrices, respectively:

$$\breve{\mathbf{u}} = \mathbf{O}\,\mathbf{R}\,\mathbf{u}.$$

The rotation matrix, **R**, is an orthonormal matrix rotating the physical signal into the sensor misaligned axes by a set of random error angles. These misalignment angles are usually very small (i.e., m°), especially if the sensor is carefully calibrated. The nonorthogonal **O** matrix may be computed by multiplying a lower-triangular matrix with a symmetric one [5]; they result in a matrix with almost unitary components on the diagonal and small random values on the lower and the upper nonsymmetric triangular blocks. The small random values are in the order of the nonorthogonality or cross-axes errors, which are commonly expressed in percent units (i.e., 5% corresponding to about 0.05 cross-axes contribution).

Output saturation, temporal discretization, and latencies

Sensor's outputs are also characterized by saturation, temporal discretization, and time delay, or latency, with respect to the physical input.

Output saturation is simply related to the maximum dynamic range of the sensor: an input value larger (or smaller) than the maximum (or minimum) range saturates the sensor's output. It can be modeled by including a saturation in the sensor model. Output saturation should not create problems in GNC application, since sensors are usually selected to have a greater dynamic range with respect to the expected dynamic ranges of the measurable quantities.

Temporal discretization of the output is due to the discrete time operations of a real GNC system, which interrogates the sensor at discrete intervals of time. Thus, this phenomenon can be modeled by maintaining a constant output between two discrete time intervals. Temporal discretization frequency of the measurements shall be compatible with the intended operating frequency of the GNC.

Latency is due to the internal processing, and it is typically quantified with ground testing. This effect can be modeled including a delay in the sensor model, and it is particularly relevant for applications with stringent synchronization requirements or with many sensors sampled at different instants of time.

Fig. 6.10 shows common model blocks used to represent these effects.

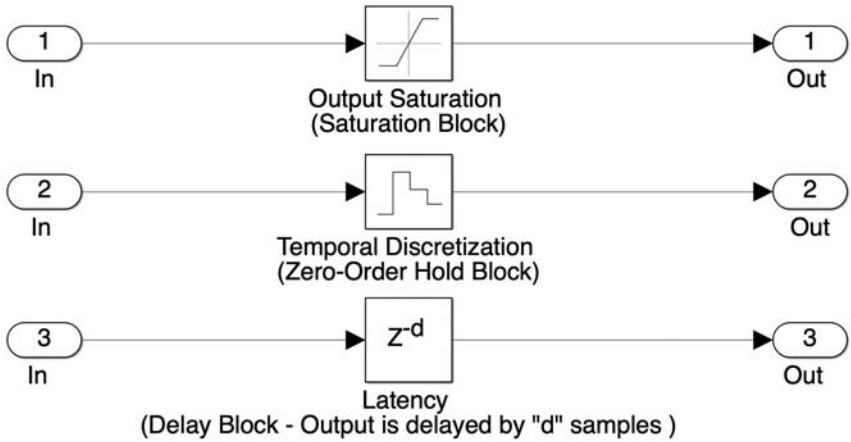


Figure 6.10 Latency, output saturation, and temporal discretization.

Sensor faults

Sensor faults have a critical impact on the GNC system. A faulty sensing unit may introduce wrong measurements in the GNC loop, leading to navigation errors bringing the system off the reference state. Similarly, the loss of a sensor may bring the spacecraft in a critical error state, where the complete functionality of the spacecraft may be not guaranteed. For these reasons, the GNC design shall take into account possible sensor faults, and it shall comply with unexpected on-board failures. Ideally, these undesired events shall be detected by the GNC itself, which shall also react to minimize their impact on the whole spacecraft. These points will be addressed in Chapter 11 — FDIR Development Approaches in Space Systems, but here a brief summary of the most common typologies of sensor faults is reported.

Sensor faults are classified and listed according to their impact on the sensor's output [6,7]. The most common are:

- *Spike fault.* Spikes are observed in the output of the sensor.
- *Erratic fault.* Noise of the sensor output significantly increases above the usual value.
- *Drift fault.* The output of the sensor keeps increasing or decreasing from nominal state.
- *Hardover/bias fault.* The output of the sensor has a sudden increase or decrease with respect to the nominal state.

- *Data loss fault*. The output of the sensor has time gaps when no data are available.
- *Stuck fault*. The sensor's output gets stuck at a fixed value.

Fig. 6.11 reports their typical appearance in a generic sensor's output.

It shall be noted that sensor models shall be also capable to reproduce the possible faults that may occur while on orbit. Hence, sensor faults need to be simulated in order to reproduce some failure scenarios that resembles real-life faults and failures by utilizing the concept of fault injection into the simulated system.

Orbit sensors

Orbit sensors are used to measure position and velocity of a satellite by using different signals and technologies. When position and velocity measurements are available, they can be used to estimate the current orbital state and, possibly, to propagate the spacecraft position forward in time. Orbit

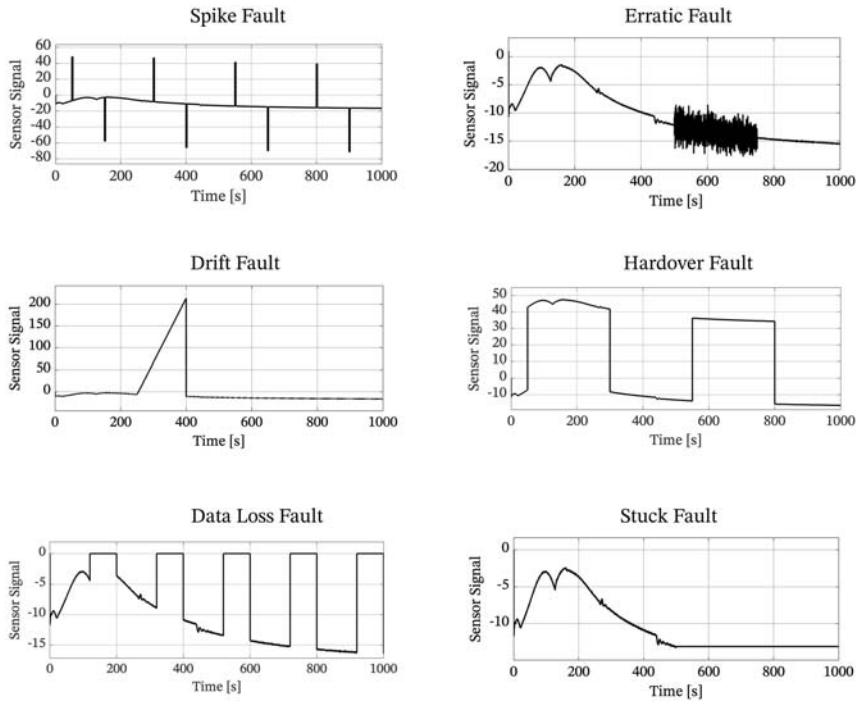


Figure 6.11 Typical sensor faults appearance, adapted from Colagrossi and Lavagna [7].

sensors belong to two main typologies: on-board or on-ground. The former enables autonomous GNC, since the measurements are directly available on the spacecraft, while the latter requires the ground support, with the upload of the measured position and velocity during the telecommunication windows.

The orbit sensors a given spacecraft can carry strongly depend on the mission and the spacecraft itself. For instance, the use of GNSS receivers is not useful for a Mars mission since GNSS satellites can only provide their signals to receivers below GNSS orbits (except for very weak signals), but it would provide a great solution for orbit determination for low Earth orbit (LEO) satellites. Therefore, when selecting the sensors suite for a given satellite, it is essential to have in mind the operational orbit and characteristics of the spacecraft. As a general consideration, on-board orbit sensors (e.g., GNSS) provide a higher level of autonomy with respect to on-ground sensors at the cost of limited applicability domain. In fact, on-board sensors are usually limited to Earth-orbit scenarios where the lower complexity of the mission and the available technologies allow for precise and autonomous orbit determination. On the contrary, the complexity to enable interplanetary orbit determination usually drives the sensor selection toward on-ground solutions. In fact, they are general and not limited to Earth scenarios, but they add operational costs, delays, and lack in autonomy.

Thus, the different types of orbit sensors can be classified depending on the autonomy and the applicability range of the navigation solutions that can be obtained by using them. In this section, on-board GNSS sensors and on-ground orbit determination with ground station networks (GSNs), along with their corresponding sensor technologies are discussed.

GNSS sensors

GNSS sensors allow a satellite or, generally speaking, any object that is below the GNSS satellites orbits (e.g., a smartphone on Earth surface) to determine its position by using the signals generated by the GNSS constellation satellites next to the ephemerides of those satellites.

GNSS basics

GNSS refers to a constellation of satellites that provides signals to receptors (that can be on land, sea, air, or even in space) with information of position and time for them to being able to determine their location. Taking into account the time it takes to signals to go from the emitting satellites in the constellation to the receptors and knowing their detailed position, it is

possible to obtain a range measurement between the emitter and the receptor. If enough signals are received, the GNSS receptor can estimate the state. There are different sets of constellations that provide global coverage, from which we can remark:

- *GPS* [8]: GPS is the oldest and more known GNSS constellation. It was designed and built, and it is currently operated and maintained by the US Department of Defense. It consists of 24 satellites, spread in 6 different orbital planes with four satellites each. The orbits semimajor axis is 26,578 km, and they have an inclination of 55°.
- *GLONASS* [9]: GLONASS is the GNSS managed by the Russian Space Forces, and it is operated by the Ministry of Defense of the Russian Federation. It consists of 21 satellites divided in 3 planes with seven satellites each. Satellites operate in nearly circular orbits with a semimajor axis of 25,510 km and an inclination angle of 64.8°.
- *GALILEO* [10]: GALILEO is the GNSS satellite constellation from the European Union and the European Space Agency with the objective of providing a very high-accuracy global positioning system under civilian control. The GALILEO constellation consists of 30 satellites in medium Earth orbit (MEO) orbit in three orbital planes with 9 satellites per orbit (plus one spare). Orbits are quasicircular with a semimajor axis of 29,600 km and 56° of inclination.
- *BEIDOU* [11]: BEIDOU is the satellite constellation from China. It consists of 35 different satellites, divided in different planes and orbit regimes. From the 35 satellites in the constellation, 8 are in geostationary regime (five in geostationary orbit [GEO] orbit and other three in inclined GEO) and the other 27 are MEO satellites that allow for complete coverage.

Apart from the global satellite positioning constellation, there are other systems that provide regional coverage in order to increase the accuracy and improve the performance of the global navigation systems, such as EGNOS or QZSS.

EGNOS is a European regional satellite-based augmentation system that uses GNSS measurements taken by accurately located reference stations deployed across Europe to improve the reliability and the accuracy of the GNSS positioning data [10].

QZSS or "Quasi-Zenith Satellite System" is a regional augmentation system for the Asia–Pacific region, especially focused on Japan, which is the developing country of the system. The QZSS system consists of 4

satellites in inclined GEOs that allow a complete regional coverage and enhancement of GNSS [12].

GNSS signals

The main objective of a GNSS satellite is to transmit a signal that is then received by a user and interpreted to extract position and velocity information. The transmitted signal has a frequency in the radio spectrum of about 1.2 and 1.6 GHz (L1 = 1.575 GHz, L2 = 1.243 GHz, L5 = 1.176 GHz) with a peculiar code modulation. In fact, pseudorandom noise code modulation is used on the harmonic radio wave (carrier). As the name suggests, this modulation has no evident pattern and it consists of a defined random sequence of zeros and ones, repeated at a given time interval and it serves as ranging code. Furthermore, a broadcast navigation message is transmitted along with the ranging code but at a lower rate. This navigation message provides information on the orbit and clocks of the transmitting satellites. More information can be found in Ref. [13].

The receiving user can interpret the GNSS signal data and extract three types of measurements:

- Pseudorange: given the time of the receiver clock at signal reception t_R and the time of satellite clock at signal transmission t_T, their difference scaled by the speed of light c can be used to construct a range measurement:

$$P = c(t_R - t_T)c$$

- Carrier phase: a direct measurement of the signal phase can be used as more precise measurement. First, the code modulation has to be removed to recover raw phase, then, the phase should be tracked to record the number of cycles. Phase information is usually more precise than pseudorange, but its initial value generates an integer ambiguity N_R. If the phase tracking is lost, the carrier phase measurements exhibit a cycle slip. A simple measurement equation considering the phase of the receiver Φ_R and the phase of the transmitted signal Φ_T is:

$$\Phi = \Phi_R(t_R) \; - \; \Phi_T(t_R) \; + \; N_R^T$$

- Doppler: the well-known Doppler effect causes a change in the received frequency. A direct measure of this change provides information on the line-of-sight velocity.

In this chapter, a simple introduction to GNSS signals is provided. For additional details on the GNSS measurements modeling, including all the relevant error effects, please refer to Chapter 9 — Navigation.

GNSS receivers

A GNSS receiver is typically mounted in LEO satellites to perform orbit determination. The GNSS receiver can provide position, velocity, and timing data (among others, such as pseudorange and carrier phase information depending on the specific receiver) that allows tracking the satellite state along the orbit. Satellites in other orbital regimes such as MEO or GEO can make use of GNSS receivers [14], and there have been different projects to study the possibility of using GNSS signals to perform navigation around the Moon, even though the precision is not as high as the one for receivers in LEO orbit and below [15].

The GNSS receivers incorporate an antenna to get the signal provided by the different GPS satellites. In principle, having three satellites in view would be enough for the receiver to determine the exact position in three-dimensional (3D) space, see Fig. 6.12. The received signals include

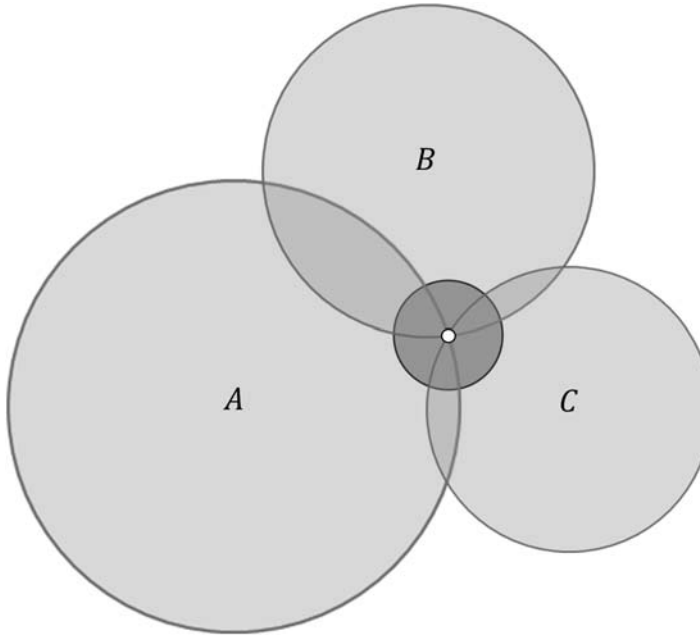


Figure 6.12 Triangulation scheme from three different satellites.

the exact timestamp of when they were generated. Therefore, by considering the generation time and the reception one, it is possible to determine the distance between the GNSS satellite that transmitted the signal and the receiver. If at least three satellites are in view of the receiver, by means of triangulation, it is possible to determine the position of the satellite with a few meters error. In modern GNSS navigation, a minimum of four satellites is usually considered to perform orbit determination.

It is important to mention that, in order for a spacecraft to being able to determine the distance with respect to the GNSS satellite, it is mandatory to know the position of the GNSS satellite itself. A set of ground stations is constantly monitoring and obtaining orbit determination solutions of the GNSS satellites. These orbital data are stored by using numerical parameters, and they are available on the GNSS satellites in the GNSS ephemerides and almanacs data. The ephemerides are broadcasted by the GNSS satellites in near real time, and they are updated at a frequency of few hours. The ephemerides allow the receiver to determine its position with respect to the satellite of the constellation in view.

However, there is an extra factor to be taken into account when determining the position from GNSS satellites: clock errors. GNSS satellites are equipped with high precision atomic clocks to maintain tracking of time, but receivers cannot be equipped with these types of clocks, since it would make them extremely expensive. For this reason, it is necessary to have a fourth satellite in view of the receiver to solve for the extra variable, time, and provide with an accurate solution of the localization problem.

GNSS accuracy

The accuracy of GNSS receivers is strongly related to the orbital regime of the spacecraft. For LEO satellites, the use of GPS can provide orbital determination solutions with errors in the order of few meters. For MEO, high Earth orbit (HEO), and GEO satellites, the accuracy is reduced up to tens of meters, and for GNSS-based orbit determination around the Moon, accuracy is in the order of hundreds of meters or even few km, depending on the geometry of the mission and the characteristics of the spacecraft. A summary of the expected accuracy is reported in Table 6.1.

Multiconstellation GNSS receivers

Multiconstellation GNSS receivers can get and process signals coming from different GNSS constellations (i.e., GPS, GALILEO, GLONASS, etc.) to determine the spacecraft state. A multiconstellation GNSS receiver has the

Table 6.1 Typical GNSS accuracy.

Scenario	GNSS accuracy—Order of magnitude
LEO	10^0 m
MEO–HEO–GEO	10^2 m
Moon	$10^3 - 10^4$ m

advantage of having a greater number of satellites in view with respect to a single constellation receiver. With the extra information provided by these new satellites, the solution obtained is not only more accurate but also more robust, since it is able to provide information even if the signal from a certain satellite is blocked by the environment (e.g., parts of the spacecraft blocking the antennas line of sight, Earth, etc.).

As opposed to older single constellation GNSS receivers (e.g., GPS receivers), in which four satellites in view are required to provide a solution, the multiconstellation receivers require five satellites to be in view. Namely, four of them are required to solve the position-time problem, and the extra satellite is needed to calculate the time difference between two constellations.

Table 6.2 summarizes the strengths and weaknesses of GNSS sensors for spacecraft orbit determination. As already discussed, GNSS offers high-accuracy measurements and autonomy capabilities. Furthermore, the development for Earth-base applications offers several solutions and a consolidated hardware design. On the contrary, the position of the GNSS constellation satellites poses an intrinsic limit to the applicability of such technology to orbits higher than GEO band.

GNSS sensor model

GNSS sensor models are based on the Earth-centered inertial (ECI) to Earth-centered Earth-fixed (ECEF) reference frame conversion. The true

Table 6.2 GNSS strengths and weaknesses.

Strengths	Weaknesses
High accuracy	Limited applicability regions (i.e., GNSS constellations shall be in view)
Extensive development	
Consolidated hardware design	
Possibility of autonomous on-board use	

inertial position and velocity, generated in an orbital propagator, are first affected by a digital white noise and a drift brown noise, representative for the internal GNSS signal electronic processing. Then, they are converted in the Earth-fixed reference, which is the GNSS reference frame. A clock model is also present, accounting for internal clock errors. All these quantities are delayed by a variable time latency and sent to output. The velocity measurement is typically further delayed with respect to the position one; this effect is included in the model as a dedicated velocity latency. An example GNSS sensor model implementation in MATLAB/Simulink is shown in Fig. 6.13.

Ground-based orbit determination

Ground-based orbit determination techniques are based on measurements performed by ground stations on the spacecraft. This method usually relies on the use of GSN, a set of ground-based antennas transmitting radio frequency signals to the spacecraft. The most common measurements derived from the processing of these signals are:

- *Range:* a measure of the range between a GSN station and the user spacecraft. Similarly, to GNSS measurements, the range is obtained considering the time it takes for a radio signal to travel from the GSN station to the spacecraft and back to the station.
- *Doppler:* it represents a direct measurement of line-of-sight velocity of a spacecraft relative to a tracking antenna. Please note that doppler measurements don't provide any information on position or velocity normal to the line-of-sight [16]. Furthermore, the radio signals can be processed in different ways:

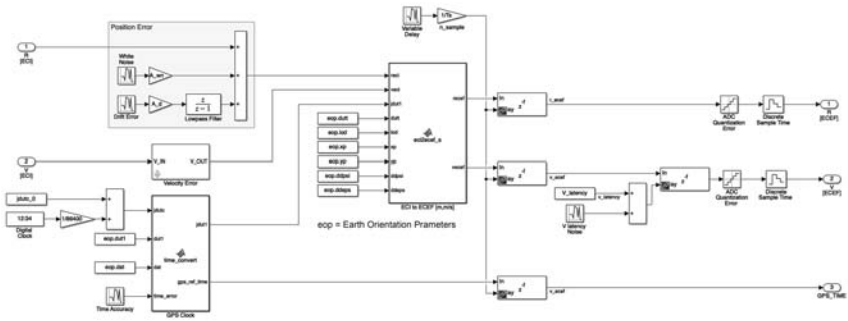


Figure 6.13 GNSS sensor model.

- *One-way:* the spacecraft generates a downlink signal that is transmitted to the GSN. The frequency of the signal is then compared, on ground, against a locally generated frequency.
- *Two-way:* the GSN transmits a signal to the spacecraft. Then, the spacecraft generates and downlinks a signal with a phase coherent with the received signal. Finally, the GSN compares the received frequency with the same reference frequency from which the uplink was generated.
- *Three-way:* the spacecraft is tracked by two stations—one using a two-way mode while the other using a one-way mode.

Please, keep in mind that two-way mode is usually employed for spacecraft navigation.

The basic radiometric measurements provided by the GSN can be also augmented by exploiting the following measurements:

- *Very Long Baseline Interferometry (VLBI)*—Δ-DOR: even if it is not considered as a basic measurement obtained by processing a radio signal, GSN has provided VLBI data since 1980 [17]. The concept of VLBI (or Δ-DOR) is to measure angular separation between a spacecraft and an extragalactic radio source (quasars or galactic nuclei). This is done by processing the radio waves coming from an extragalactic object (ideally close to the spacecraft) and from the spacecraft, in two tracking stations that have a large physical separation. By differentiating the delays of the two incoming signals, the angular separation between the extragalactic source and the spacecraft, in the direction perpendicular to the baseline, can be obtained.
- *Optical data:* usually obtained from a scientific camera available on-board, optical data provide a measure of the direction of a vector from the spacecraft to a target body (e.g., planet, small body). The measured direction can be combined with other range measurements to obtain a complete 3D position of the spacecraft. Angular accuracy of 10 μrad is expected.
- *Altimetry:* an altimeter provides a measure of the distance between the spacecraft and a target body (e.g., planet, small body). When combined with optical data, altimetry measurements can allow for a complete 3D positioning of the spacecraft. Its operational envelope, however, is limited to distances very close to target body, and, therefore, it is only marginally useful for ground-based spacecraft orbit determination.

In the case of on-ground orbit sensors, the needed infrastructure is usually divided into ground and space segments. This is different with respect to on-board sensors, as the GNSS receivers, which allow obtaining an orbit determination solution without involving ground in the loop.

Ground segment

The ground segment englobes all the ground-based elements of the infrastructure required to perform orbit determination, such as ground stations, mission control, ground network connecting the elements or terminals.

GSNs, such as Deep Space Network from NASA, or ESTRACK from ESA, provide links between the satellites in orbit and the teams on ground. They are in charge of sending and receiving signals with command, information of spacecraft status as well as signals dedicated to navigation purposes [18]. Moreover, the ground segment also includes all the navigation functions to perform orbit determination and process the data to obtain spacecraft position and velocity estimates.

Space segment

Apart from the ground stations on Earth, extra elements are required, in this case on the spacecraft itself, to perform the orbit determination process. Typically, this element is the transponder for two-way range observables. The transponder on-board receives the signals generated by the ground stations, demodulates it, and transmitted it back to ground, using phase modulation on the downlink carrier, making it a signal coherent with the uplink but shifted in frequency.

Ground-based orbit determination accuracy

The typical accuracy of the presented ground-based measurements is reported in Table 6.3.

VLBI clearly improves the overall orbit determination accuracy based only on range and doppler measurements. Optical data are not equally accurate, but provide an additional measurement to retrieve a 3D position. Finally, the use of altimeter improves significantly the position accuracy, but it is limited to close-proximity operations.

Finally, Table 6.4 summarizes strengths and weaknesses of the sensors for ground-based spacecraft orbit determination. As discussed before, ground-based sensors offer high-accuracy measurements in a wide range of possible applicative scenario. However, these solutions imply limited autonomy and communication delays along with a significant operational cost.

Table 6.3 Typical ground-based orbit determination accuracy.

Measurement	Measurement accuracy—Order of magnitude
Range	10^{0} m
Doppler	$10^{-3}\frac{\text{m}}{\text{s}}$
VLBI	10^{1} nrad
Optical data	10^{4} nrad
Altimetry	10^{-2} m—dependent on the relative distance

Table 6.4 Ground-based OD strengths and weaknesses.

Strengths	Weaknesses
High accuracy	Limited autonomy
Extensive applicability	High cost
	Communication delays

Attitude sensors

Attitude sensors are used to measure reference directions in the spacecraft body frame, which are then compared with the on-board stored definition of the same directions in the inertial frame. In this way, it is possible to determine the attitude of the spacecraft body frame with respect to the inertial one, as will be discussed in the attitude determination section of Chapter 9 — Navigation.

Many directions could be taken as references, even if the most common attitude sensors measure well-known environmental quantities: star positions, Earth center or limb, Sun direction, or Earth's magnetic field. In fact, there exist four types of sensors that are typically used in attitude determination and control. They are listed below in an approximated order of accuracy:

- Magnetometers.
- Sun sensors.
- Horizon sensors.
- Star sensors.

The following sections describe the different technologies available for each type of sensors, together with the main characteristics, advantages, and drawbacks.

Magnetometers

Magnetometers are a very common type of attitude sensors, flying on most of the satellites in LEO, while they are rarely used in MEO and practically nonpresent in GEO. Similarly, they are not used for interplanetary orbits. This is due to the decreasing of the Earth's internal magnetic field with the square of distance from its center. Thus, for spacecraft orbiting at larger distance from the Earth, the magnetic field is less convenient for both attitude estimation, due to the increased relative error of the sensor, and control, due to the lower ratio of torque over actuator mass. Moreover, magnetometers are also rarely employed in missions around other celestial bodies with a strong magnetic field, since the knowledge of these environments is too poor to have a reliable exploitation of magnetic measurements and torques.

Magnetometers do not provide a direct attitude information or estimation, but they only measure the surrounding magnetic field in magnitude and direction. Since this is a single vector, the satellite state is not fully observable, and it is necessary to combine this measurement with other sources of information to derive the complete attitude. This can be done in several ways, usually using a subset of the below options:

- Other attitude sensor measurements, as Earth sensors or Sun sensors.
- A magnetic field model provided the satellite position is known, either by orbit propagation or estimation using a GNSS [19,20].
- Gyro measurements, together with attitude propagation and other observation techniques.

Beside for estimation of the attitude, the magnetometers are used as well to properly drive magnetic actuators (i.e., magnetorquers in Chapter 7 — Actuators), since the magnetic torque is the function of both the satellite magnetic dipole and the local magnetic field. Hence, a spacecraft with magnetic actuation shall have on-board measurements of the surrounding magnetic field. Note that an on-board stored magnetic field model is not needed to compute magnetic torque commands.

Another use of magnetometers is to provide a coarse angular rate estimation by time derivation of the computed magnetic field unit vector [21]. This operation is usually performed in combination with other sensors, again due to the observability problem of the satellite state.

Magnetometers for attitude measurements are those of vector type, which are capable to measure the vector components of a magnetic field. In fact, they are typically composed by at least three orthogonal single-

axis magnetometers, as in Fig. 6.14. The most common typologies of vector magnetometers for space applications are:

- Magnetoresistive sensors, made of a nickel–iron alloy with a high magnetic permeability, whose electrical resistance varies with a change in magnetic field. They have a very quick response time, but their accuracy is moderate, ranging from 1° to few degrees in direction, and 10 to 50 nT of resolution.
- Fluxgate magnetometers, made of magnetically susceptible cores wrapped by two coils of wire as shown in Fig. 6.15. One coil is connected to an alternating electric current, which drives the cores through opposite alternating cycles of magnetic saturation. If no external magnetic field is present, the fields in the two cores cancel each other and the induced field in the secondary sensing coil would be zero. However, when the cores are exposed to a background field, the one in alignment with the external field has an increased magnetic induction. The contrary happens in the core opposite to the background field. In this way, an alternating electric signal could be induced in the second coil to be measured by a detector. Specifically, the voltage at the secondary coil is the derivative of the flux of the magnetic field vector. In order to achieve meaningful output, the two cores shall reach magnetic saturation during the

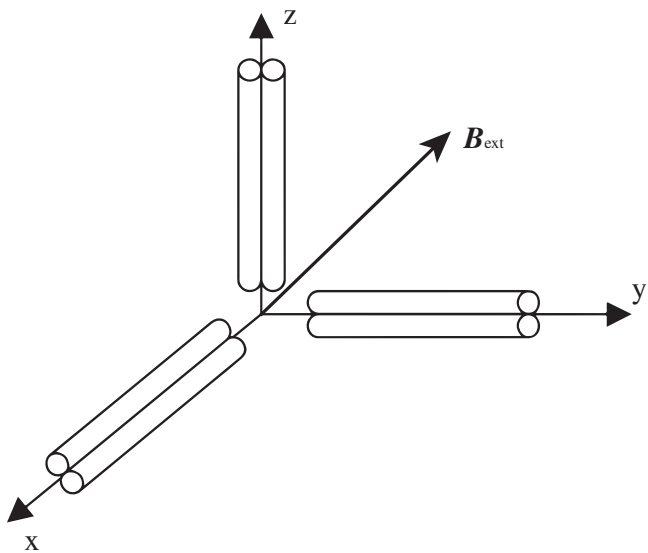


Figure 6.14 Three-axis magnetometer.

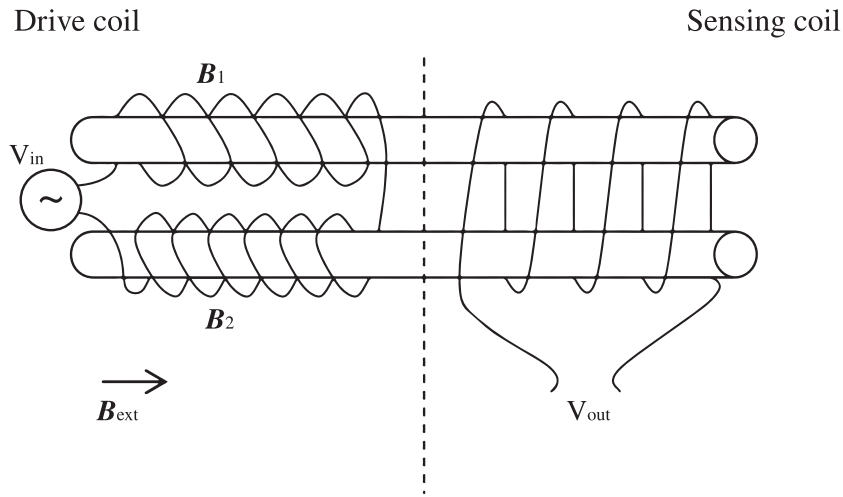


Figure 6.15 Fluxgate magnetometer.

alternating cycle. Hence, the driving current in the primary coil shall consider the expected value of the external magnetic field. Fluxgate magnetometers are the most common for spacecraft applications, since they are relatively small, lightweight, with no moving parts, and inexpensive. They have a quick response time and a good accuracy in the order of 0.1° in direction, with a resolution of 0.1 to 10 nT.

- Microelectromechanical systems (MEMSs) magnetometers for very small-scale applications. This typology of sensor is usually operated by detecting the effects of the Lorentz force or the quantum mechanics effects of the electron tunneling. These sensors are extremely light and power efficient, but they are less accurate with respect to fluxgate magnetometers, with an accuracy in the order of 1°.

Although the accuracy of the measured magnetic field direction is not incredibly high (e.g., ranging approximately from 0.1° to several degrees, depending on the technology used), they are very reliable sensors. Thus, they are typically used in magnetic angular rate damping safe modes (cfr. Chapter 14 — Applicative GNC Cases and Examples). Another reason for their popularity among GNC systems is that they are quite affordable in terms of costs and on-board resources, compared to other satellite equipment.

The magnetic field measured by the sensor is the composition of Earth magnetic field and the magnetic field generated by the satellite itself. For

this reason, magnetometers accommodation must be properly studied in order to guarantee a level of magnetic cleanliness better or comparable to the sensor performance. Magnetometers can be accommodated both internally and externally to the satellite body, possibly on a boom to increase the distance between the sensor and the satellite body. This is to take advantage of the $1/r^3$ falloff of a magnetic dipole field.

The internal magnetometers accommodation shall be done in a part of the satellite distant from magnetic sources (e.g., attitude control torquers, dipoles, ferromagnetic materials, power lines, current loops in solar arrays, electric motors, etc.). As a rule of thumb, a local magnetic field has to be avoided comparable to the magnetometer bias, in order not to reduce the sensor accuracy. As a mitigation action, it is possible to calibrate the sensor to compensate for this error. However, it is difficult to estimate the actual disturbance while on orbit, since it can be different from what is measured on ground.

If the magnetometer is mounted externally, the sensor is exposed to a more severe space environment. In this case, the main criticalities are the wider range of temperature to be coped with and the increased radiation flux. The shielding to achieve a reduced level of radiation is typically not a problem for analog space qualified sensors, while it may be more problematic for digital sensors or, in general, for commercial components.

Special care is needed to guarantee the compatibility of magnetometers measurement to the use of magnetorquers, which produce a strong and variable local magnetic field, as described in Chapter 7 — Actuators. Indeed, when magnetorquers are activated, they generate a magnetic dipole that can easily saturate the magnetometer (i.e., the generated magnetic field is higher than the magnetometer full scale). This can be solved by scheduling the use of magnetometers and magnetorquers at different times, such that the magnetorquers are commanded with a certain duty cycle and the magnetometers measurements are not acquired at the same time. Again, as a rule of thumb, the time distance between the magnetorquers actuation and the magnetic measurements should be at least three times the magnetorquers characteristic time to allow enough magnetic dipole attenuation.

If the saturation of the measurements is not a problem, the known local magnetic field generated by the magnetorquers can be compensated in the magnetometer's output.

Table 6.5 summarizes the strengths and weaknesses of magnetometers.

Table 6.5 Magnetometers strengths and weaknesses.

Strengths	Weaknesses
Reliable	Poor accuracy (0.5−5° approximately)
Affordable	Measurement influenced by satellite magnetic cleanliness
Measurement always available on the orbit	Not suited for orbits above LEO
Used both for attitude estimation and magnetic torquers driving	No direct attitude information

Sun sensors

A Sun sensor is a device that senses the direction of the Sun with respect to the sensor reference frame.

The Sun sensor is operated based on the entry of light into a thin slit or a pinhole on top of a chamber whose bottom part is lined with a group of light-sensitive cells. The chamber casts an image of the aperture on the detector. The cells at the bottom measure the distance of the image from a centerline and determine the refraction angle by using the chamber height.

The receptive cells convert the incoming photons into electrons and, thus, voltages, which are in turn converted into a digital signal. When two groups of light-sensitive cells are placed perpendicular to each other, the direction of the Sun with reference to the sensor axes, and consequently the spacecraft body frame, can be calculated.

The design of a Sun sensor system for any spacecraft application must provide acceptable physical characteristics, meet performance specifications, and cope with interference sources and environmental conditions while minimizing penalties to spacecraft design and performance. Physical characteristics include size, weight, and power requirements. The major performance parameters are accuracy, stability of the measurement, field of view, transfer function characteristics, and resolution. Interference sources include sunlight reflected from Earth and other planetary bodies, reflection from spacecraft surfaces, and spacecraft electromagnetic fields [22]. In particular, Earth albedo determines a significant disturbance in the Sun sensor measurements, and it shall be calibrated in the GNC algorithms to achieve the full sensor performance. Hence, GNC functions shall be capable to estimate the incoming light radiation from the Earth to be subtracted from the sensor measurements.

The purpose of this chapter is to highlight the most important characteristic that a GNC engineer needs to consider when designing a system entailing Sun sensors. Here, we limit the description to sensors that indicate the orientation of the Sun with respect to a known reference frame. The orientation is sensed by detecting the intensity difference between radiation arriving from the solid angle subtended by the Sun boundaries and the rest of the radiation reaching the sensor from the adjacent regions surrounding the sensor's field of view.

Sun sensors are usually classified into three types:

1. Analog sensor whose output signal is a continuous function of the angle of incidence.
   a. Coarse Sun sensors (CSSs).
   b. Fine Sun sensors (FSSs).
2. Digital sensor that produces encoded discrete output of the function of the angle of incidence.
3. Sun presence sensor that provides a constant output signal when the sun is in field of view.

Analog sun sensors

The analog Sun sensors deliver a continuous function based on the incident radiation. The analog sensors are furtherly classified in CSSs and FSSs.

Coarse sun sensors

These types of analog sensors measure the amount of incident light without using windows nor projections. Incident light flux is proportional to the cosine of the angle between the sun and the normal vector of the light-sensitive cell, which allows us to calculate the angle of incidence of the solar vector. With reference to Fig. 6.16, the equation simply reads:

$$J = J_s \cos(\alpha)$$

where J is the solar intensity at the sensor, J_s is the Sun flux light constant, and α is the incidence angle between the light direction and the normal to the sensor.

Solar sensors of this type only use one photodiode, and the electric current they create is used to deduce the angle of incidence. At this point, it is insightful to think of the solar panels as a particular type of CSSs because of their capability to generate current following the cosine law mentioned before.

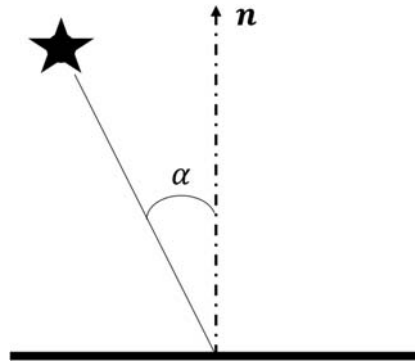


Figure 6.16 Lambert cosine law.

If we assume the Sun as a light point source identified by the $\boldsymbol{s_b}$ unit vector from the spacecraft to the Sun, the output of a CSS mounted on the face A, whose normal is identified by $\boldsymbol{n}_A$, is [23]:

$$I_A = \begin{cases} I_{\max}(\boldsymbol{n}_A \cdot \boldsymbol{s_b}) \text{ for } \boldsymbol{n}_A \cdot \boldsymbol{s_b} > 0 \\ 0 \quad \text{for } \boldsymbol{n}_A \cdot \boldsymbol{s_b} < 0 \end{cases}$$

where I is the CSS output current. Let us now assume that we have another sensor on the other side of the spacecraft, namely on face -A, whose normal is identified by $-\boldsymbol{n}_A$, then:

$$I_{-A} = \begin{cases} I_{max,-A}(\boldsymbol{n}_{-A} \cdot \boldsymbol{s_b}) = -I_{max}(\boldsymbol{n}_A \cdot \boldsymbol{s_b}) \text{ for } -\boldsymbol{n}_A \cdot \boldsymbol{s_b} > 0 \\ 0 \quad \text{for } -\boldsymbol{n}_A \cdot \boldsymbol{s_b} < 0 \end{cases}$$

Hence, we can write:

$$I_A - I_{-A} = I_{max}(\boldsymbol{n}_A \cdot \boldsymbol{s_b}), \forall \boldsymbol{n}_A \cdot \boldsymbol{s_b}$$

Adapting the exhaustive example in Ref. [23], given the above relationships, if we equip the spacecraft with at least 6 not coplanar Sun sensors, $\pm \boldsymbol{n}_i, \pm \boldsymbol{n}_j, \pm \boldsymbol{n}_k$:

$$\begin{bmatrix} I_i - I_{-i} \\ I_j - I_{-j} \\ I_k - I_{-k} \end{bmatrix} = I_{max} \begin{bmatrix} \boldsymbol{n}_i \cdot \boldsymbol{s_b} \\ \boldsymbol{n}_j \cdot \boldsymbol{s_b} \\ \boldsymbol{n}_k \cdot \boldsymbol{s_b} \end{bmatrix} = I_{max} \begin{bmatrix} \boldsymbol{n}_i^T \\ \boldsymbol{n}_j^T \\ \boldsymbol{n}_k^T \end{bmatrix} \boldsymbol{s_b}$$

The Sun unit vector can be computed as:

$$\boldsymbol{s_b} = \frac{1}{I_{max}} \begin{bmatrix} \boldsymbol{n}_i^T \\ \boldsymbol{n}_j^T \\ \boldsymbol{n}_k^T \end{bmatrix}^{-1} \begin{bmatrix} I_i - I_{-i} \\ I_j - I_{-j} \\ I_k - I_{-k} \end{bmatrix}$$

These sensors are very sensitive to variations in temperature, which can cause differences in the current generated, and the estimated angle is directly proportional to this current, leading to imprecise readings of the angle of the solar vector. They are not very precise when the incident angle is close to perpendicular. For this and other reasons, coarse sensors are not as accurate as fine sensors.

Fine sun sensors

In Fig. 6.17, which represents a single-axis fine analog Sun sensor, two photosensitive elements are placed close to each other. The Sun rays are collimated or deviated to form an apparent image on the elements. Each element outputs an electrical current of different intensity based on the Sun direction. The difference between the current outputs developed across the two detector elements yields the sensor output. Whenever the two currents are equal, meaning that the Sun illumination on the two elements is equal, the sensor output is zero and the sensor reached the null point, with the Sun perpendicular to the sensor.

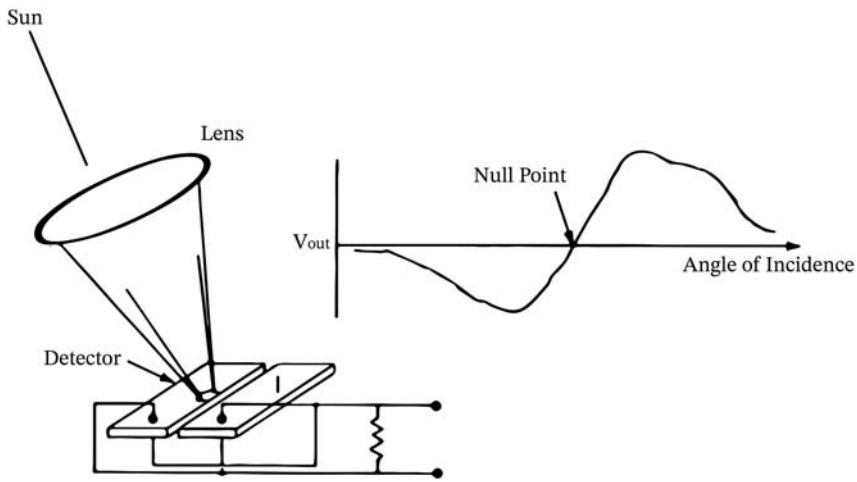


Figure 6.17 Analog Sun sensor.

Digital Sun sensors

In Fig. 6.18, the working principle of a digital Sun sensor is schematized. The Sun is imaged as a line across an array of separate elements. Each element produces a 1 or 0 binary bit in the multichannel output, depending on whether light reaches the element through the mask and whether the sensor's output in each channel exceeds threshold values established in associated circuitry. The binary number assigned to the channel identifies the position of the light imaged on the array of elements. The increased number of photosensitive units makes them more sensitive, increasing the accuracy of the sensor. Modern digital Sun sensors exploit multipixel optical detectors to further increase the sensor accuracy.

As a GNC engineer, an important step is the modeling of the sensors as a mathematical equation. We have seen that the Sun's relative position is sensed with respect to the spacecraft body frame. This unit vector, called Sun measurement vector, is output by the sensor in the body reference frame. To determine the spacecraft attitude, it is necessary to determine the rotation between a known reference frame and the body reference frame ${}_b\mathbf{A}_i$.

Thus, the sensor can be modeled, according to sensor modeling section, as:

$$\mathbf{s}^b(t) = {}_b\mathbf{A}_i\mathbf{s}^i(t) + \boldsymbol{\varepsilon}(t),$$

where $\boldsymbol{s}_b$ is the Sun vector in the body frame, and $\boldsymbol{s}_I$ is the Sun vector in the inertial reference frame, which is typically adopted as reference and $\boldsymbol{\varepsilon}$ the expected sensor errors. At this point, it becomes clear that an attitude estimation algorithm that uses the Sun sensors measurements requires the

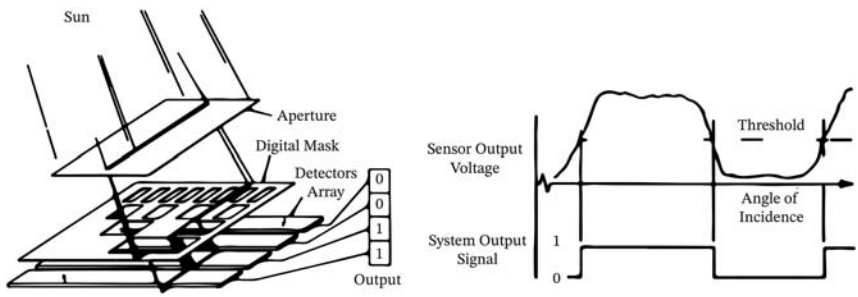


Figure 6.18 Digital Sun sensor.

knowledge of the position of the Sun with respect to a known reference frame, which is typically an inertial one. Such information can be retrieved from Sun ephemerides or from a simplified model of the Sun motion with respect to the spacecraft orbital reference frame [24].

To summarize, the most important qualitative characteristics of the various types of Sun sensors is reported in Table 6.6.

In general, Sun sensors can be quite accurate, i.e., roughly $<0.01°$, but high-resolution FSSs can reach up to 1 arcmin resolution. Unfortunately, especially for LEO orbits, they are subject to occultation during eclipse times. For this reason, when designing a system based on Sun sensors, it is remarkably important to consider the shortage provoked by the loss of this signal without degrading the GNC performance (e.g., using an alternative sensor or propagating the attitude state with a gyroscope). Furthermore, Sun sensors need clear fields of view.

Table 6.7 summarizes the strengths and weaknesses of Sun sensors.

Sun presence sensors

Analog and digital Sun sensors are more complex compared to the Sun presence sensor, which delivers a simple binary output. Indeed, Sun presence sensors are only capable to detect if the Sun is in front of the sensor, within a certain angle, or not. This sensor typology may be helpful to implement fast Sun searching or Sun avoidance algorithms.

Sun sensor model

Sun sensor models are based on a modeling approach rotating the ideal spacecraft-to-Sun unit vector in the sensor reference frame, aligned with

Table 6.6 Types of Sun sensors.

Type		Features
Coarse Sun Sensors	It only provides the relative intensity of the incident light rays according to the cosine law.	
Fine Sun sensors	Digital	• Digital output • No analog detection circuitry • Less noise • More power consumption
	Analog	• Analog output • Less power consumption

Table 6.7 Sun sensors strengths and weaknesses.

Strengths	Weaknesses
Reliable	Not working during eclipses
Affordable	No direct attitude information
It can be used in a large portion of the Solar System	Affected by perturbing light sources
Good accuracy	

its line of sight. In the case the Sun is within the sensor's field of view, the Sun vector is processed adding observation errors affecting both the Sun angles and the luminous intensity. The angles are affected by brown electronic noise, distortions due to geometrical aberrations, and scale factor errors. Finally, the Sun vector is converted in the output data format, as described in the sensor datasheet, thanks to some geometrical transformations. The possible luminous disturbance introduced by the Earth's albedo is also modeled. An example digital Sun sensor model implementation in MATLAB/Simulink is shown in Fig. 6.19. A similar modeling approach can be also used to model Earth horizon sensors or Star sensors. In the latter case, the Star sensors can estimate the complete attitude of the spacecraft, by comparing the visible stars with their positions saved in a star catalog, as will be discussed in the star sensors section. Thus, the star sensor model shall also include the star catalog, the star matching process, with possible identification errors, and the final attitude reconstruction.

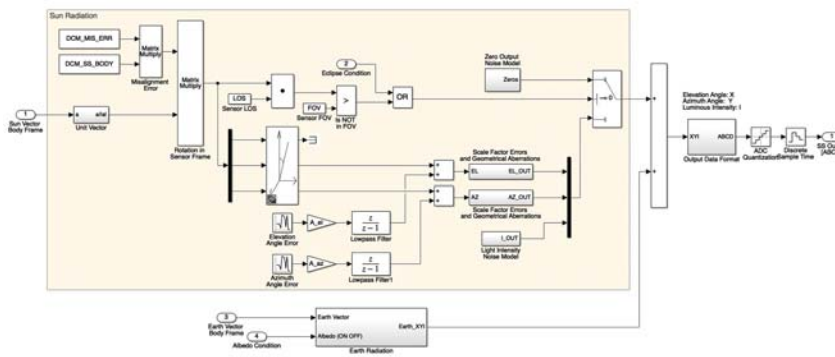


Figure 6.19 Sun sensor model.

Horizon sensors

Horizon sensors (often referred specifically to Earth sensors) are infrared devices that can detect the discontinuity in temperature between the deep space and the Earth (or planetary) atmosphere. An important task that required considerable effort during the historical development of these types of sensors is the determination of the best spectral region for defining the space-to-Earth discontinuity and providing robustness to disturbance radiation. Most frequently, sensors use the Earth's radiation in the infrared spectrum (from 2 to 30μm) and, in particular, within the narrow 14−16μm [23,25]. The reasons why horizon sensors for Earth limb detection are often centered in the above range are:

- The variation between maximum and minimum radiance is smaller compared to the visible spectral band.
- When imaging the Earth in the infrared spectrum, the terminator line on the Earth surface vanishes. This is because the difference between night and day temperature is negligible with respect to the absolute temperature scale (i.e., with reference to the absolute zero).
- Imaging in the infrared spectral band prevents the loss of signal during eclipses.

Earth presents a finite size and cannot be interpreted as a point because the approximation would just be too erroneous. Indeed, the solid angle subtended by the Earth is roughly 3.9 steradians at 500 km altitude, compared to the approximation of the Sun, which extends for nearly $7 \cdot 10^{-5}$ steradians [26]. In general, the most important parts of a horizon sensor are the infrared optical system, including the spectral filter, and the focal plane where radiance detectors are placed, e.g., thermistors or thermopiles, as show in Fig. 6.20.

Horizon sensors are classified into two distinct typologies:

1. Static sensors: they are mounted fixed on the spacecraft in predetermined body axes directions. For this reason, this kind of sensors is dedicated to spacecraft that have always (or for a large part of the time) the Earth within the sensor field of view. A typical end-to-end pipeline of the sensor processing is shown in Fig. 6.21.
2. Scanning sensors: they are composed of a moving detector, whose field-of-view (FOV) scans the Earth. Scanners are sensors that either mechanically, electronically, or passively (on a rotating vehicle) scan a large volume of object space with a scan pattern fixed relative to the sensor or vehicle [26].

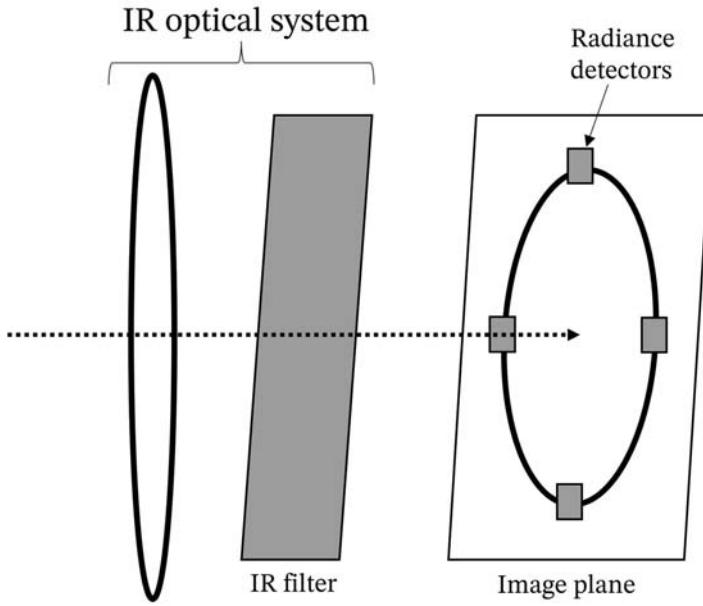


Figure 6.20 Schematic of a horizon sensor.

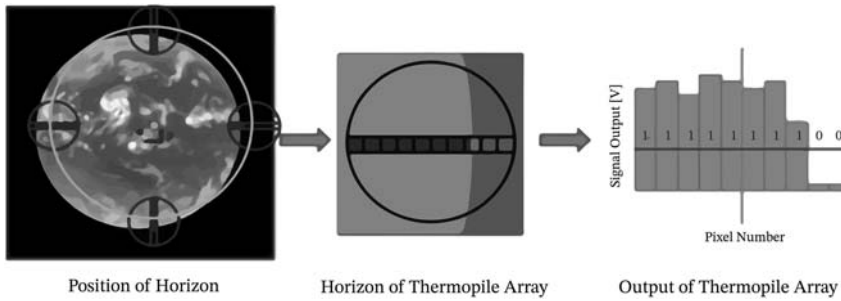


Figure 6.21 Workflow of Earth horizon sensor functioning.

The accuracy of horizon sensors ranges from 0.1 to 0.5° [23,26].

As already mentioned, the Earth sensor delivers a measurement of the relative attitude between the Earth and the spacecraft. Given the shape of the Earth (circular as a first approximation), the output of a horizon sensor inherently possesses an ambiguity in the yaw angle. In other words, the rotation around the radial axis of the comoving local-vertical-local-horizontal (LVLH) frame (cfr. Chapter 2 — Reference Frames and Planetary Models)

is not determined. Only, roll and pitch angles can be estimated by the horizon sensor. In its simplest form, the model of an Earth horizon sensor reads:

$$\breve{\mathbf{x}} = \begin{bmatrix} \phi_{b,E} \\ \theta_{b,E} \end{bmatrix} = \begin{bmatrix} \phi_{b,LVLH} \\ \theta_{b,LVLH} \end{bmatrix} + \begin{bmatrix} \alpha_{LVLH,E} \\ \beta_{LVLH,E} \end{bmatrix} + \boldsymbol{\varepsilon},$$

where $\phi_{b,LVLH}$ and $\theta_{b,LVLH}$ are the true roll and pitch attitude of the satellite, i.e., the spacecraft body frame with respect to the LVLH frame; $\alpha_{LVLH,E}$ and $\beta_{LVLH,E}$ are the angles between the Earth's horizon and the LVLH frame. Finally, $\boldsymbol{\varepsilon}$ is the measurement error. For circular orbits and under the assumption of a spherical Earth, the angles between the Earth horizon and the LVLH comoving orbital frame are constant and equal to:

$$\alpha_{LVLH,E} = \beta_{LVLH,E} = \pi - sin^{-1}\left(\frac{R_E}{R_E + h}\right)$$

where R_E is the Earth radius and h is the orbital altitude.

Table 6.8 summarizes the strengths and weaknesses of horizon sensors.

Star sensors

Star sensors (or star trackers) are devices that image the celestial sphere and determine the attitude of the spacecraft with respect to the inertial reference frame, where the positions of the stars are cataloged. Star sensors are basically optical cameras, see Fig. 6.22, which use detectors to react at the incoming light from the stars. The stars' directions are extrapolated from the image plane to the spacecraft body-fixed reference frame. Each star ephemeris is stored in a preloaded catalog; hence, the correspondence between the imaged stars and database ones delivers an estimation of the spacecraft attitude. Indeed, by measuring more independent directions, star sensors can

Table 6.8 Horizon sensors strengths and weaknesses.

Strengths	Weaknesses
Roll and pitch attitude information	It works well in the vicinity of planets or, in general, for planetary pointing mission (e.g., Earth nadir pointing)
Good accuracy	Complex calibration of spectral region
Not affected by eclipses	Not complete attitude information

completely solve the attitude problem without the need of other primary sensors. This process and the relevant determination algorithms to work out the attitude problem are detailed in Chapter 9 – Navigation.

According to Ref. [27], a star sensor comprises an imaging function, a detecting function, and a data processing function. The imaging function collects photons from objects in the field of view of the sensor and focuses them on a detecting element. This element converts the photons into an electrical signal that is then subject to some processing to produce the sensor output. International space standards [27] thoroughly define the classification and capabilities of different star sensors. Hereby, a brief summary on the most important capabilities of star sensors is reported:

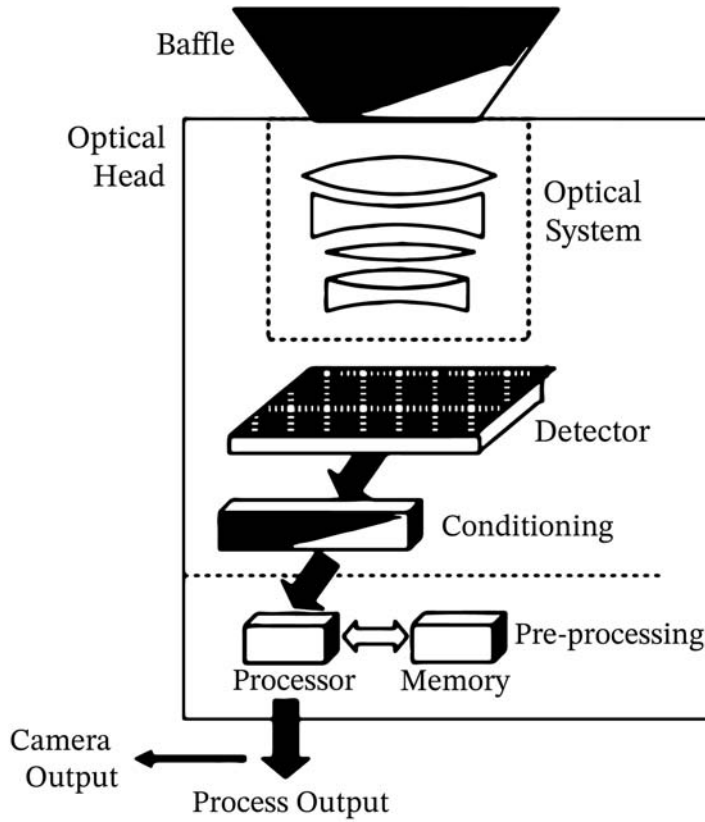


Figure 6.22 Schematic of star sensor elements [27].

- *Cartography:* this function entails the capability of delivering the star position and measurement date for each detection. Star position is identified in the sensor reference frame, whose rotation with respect to the spacecraft is known.
- *Autonomous attitude determination:* this function entails the capability of delivering the orientation of the sensor reference frame with respect to the adopted inertial reference frame. According to ECSS [27], the sensors implementing such functionality should deliver a validity flag of the determined attitude. This capability is often referred as the *lost-in-space problem* solution. Initial attitude acquisition computes at least three centroids for the brightest clusters of pixels, representative of stars. A set of descriptors for these detections groups, comprising brightness and arc lengths are generated and used in the pattern searching algorithms to match the detections with known stars.
- *Star tracking:* this function entails the capability of delivering the position of selected stars with respect to a sensor-attached reference frame. This capability does not imply to autonomously identify the stars to be tracked or explicitly identified by the unit. Indeed, the initial selection of the star images to be tracked is not autonomous and must be done offline. However, it maintains the identification of each star image and correctly updates the coordinates of each star image as it moves across the detector due to the angular motion of the sensor. In this way, it allows for faster attitude problem solutions by comparison with respect to previously known attitude states.

Stars are highly accurate references and nearly independent by the orbit position. Furthermore, they are generally available everywhere in the sky. For these reasons, star sensors are among the best attitude sensors in terms of accuracy, which falls below few arcseconds in the sensor boresight direction and slightly larger errors for rotations around the boresight axis. Nevertheless, the increased accuracy comes with considerable cost, in terms of mass and power consumption with respect to Sun and horizon sensors. Star sensors require time in solving the initial attitude problem from a lost-in-space condition, and they are generally slower than other attitude sensors. In fact, during maneuvers, there might be the need of attitude propagation with gyroscopes, since star sensors could lose the attitude solution if large slew rates are present. Moreover, star sensors are delicate since they are disturbed or

even damaged if their field of view is directed to an intense light source (e.g., the Sun or the Earth). For this reason, they are typically equipped with protecting baffles and need to be covered if they are pointed to the luminous object. Finally, their optical elements shall be protected from contaminants, such as thruster plumes.

The geometric modeling of a star sensor resembles the pinhole camera model, which will be thoroughly discussed in Chapter 9 — Navigation. For the sake of clarity, it is here reported the founding relationship of a generic point pinhole projection, as in Fig. 6.23.

Whenever a star is imaged, a block of pixels is illuminated due to the optics being slightly defocused, hence yielding a point spread function distribution rather than a unique pixel. A centroiding algorithm determines the pixel location of the star by calculating a weighted average of the pixels' brightness composing the block. The resulting detection accuracy depends on the sensor, the star brightness, the exposure time, and various errors related to the implementation of the optics [23]. For a GNC engineer, not strictly involved in the manufacturing of these sensors, it is important

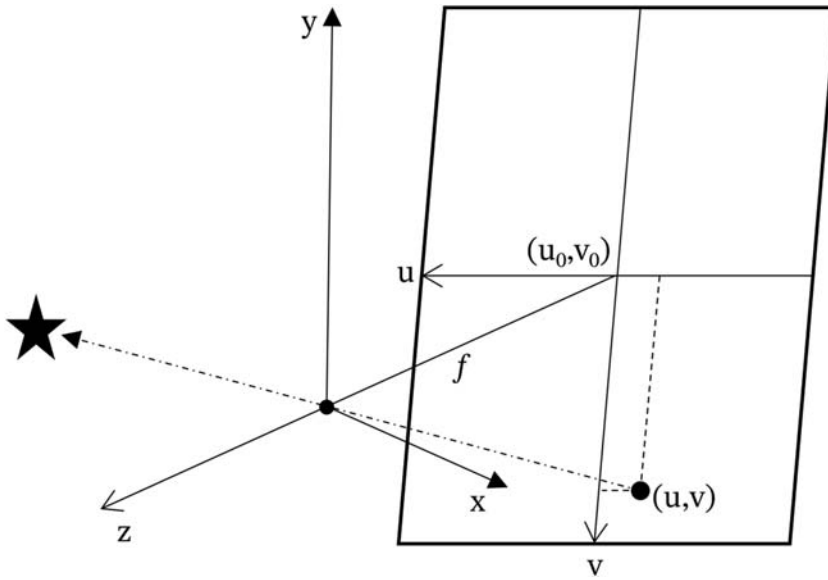


Figure 6.23 Star sensor modeling under the assumption of pinhole camera. The reference frame *x, y, z* is the sensor frame; *u,v* is the image frame and *f* is the focal length.

to obtain all the data (e.g., from the supplier) to characterize the sensor performance. A generic measured star vector **s**, namely the unit vector from the spacecraft to the identified star, can be computed with reference to Fig. 6.23 as:

$$\mathbf{s} = \frac{1}{\sqrt{\left(f^2 + (u-u_0)^2 + (v-v_0)^2\right)}} \begin{bmatrix} u-u_0 \\ v-v_0 \\ f \end{bmatrix} + \boldsymbol{\varepsilon}$$

where $\boldsymbol{\varepsilon}$ is the measurement error. The star catalog used in a star sensor is a critical variable that largely affects the accuracy of the sensor. Typically, sensors offer an end-to-end pipeline depending on the capability the sensor provides, as mentioned before (e.g., cartography, star tracking, attitude determination). Nevertheless, it is interesting to report how the number of stars influences the sensor output. To make a proper attitude determination, we may need at least N = 3/4 stars to be identified in the image. The more objects are identified, the more robust to random errors our estimation will be. The probability of finding at least N stars in the sensor field of view can be assumed as a Poisson distribution [23]:

$$P(N) = e^{-N_{avg}} \frac{N_{avg}^N}{N!}$$

where N_{avg} is the average number of stars we expect in the FOV. The average number of stars in FOV can be determined by the FOV angle and the number of catalog stars $N_{catalog}$, knowing that the size of the celestial sphere is 4π sr:

$$N_{avg} = \frac{N_{catalog} \cdot FOV^2}{4\pi}$$

This means that the probability of finding at least N stars in the FOV is influenced by the number of stars in the catalog and the instrument FOV. Given a certain value of average number of stars in FOV, the probability of finding and detecting at least N stars in the FOV is represented in Fig. 6.24.

For instance, with reference to Fig. 6.24, the probability of finding at least $N = 5$ stars with an average of $N_{avg} = 8$ stars in FOV is 90%. The number of catalog stars or the required FOV can be determined as:

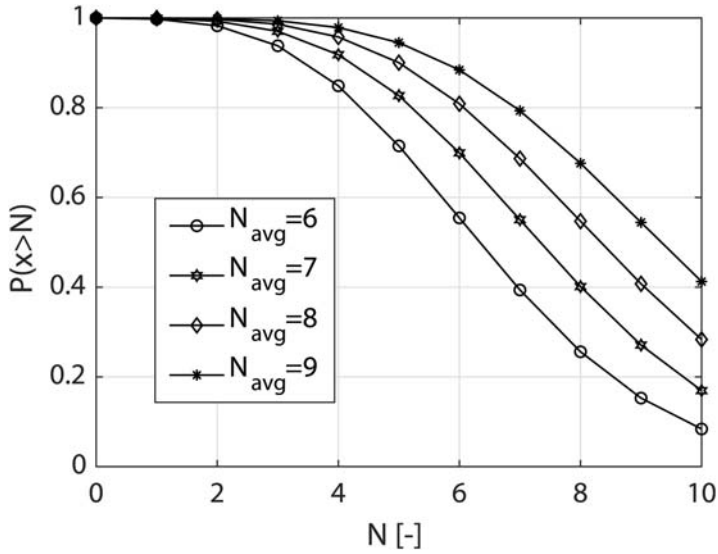


Figure 6.24 Number of detected stars in FOV for different average number of stars in FOV, linearly dependent from the number of catalog stars.

$$FOV = \sqrt{\frac{4\pi N_{\text{avg}}}{N_{\text{catalog}}}}$$

$$N_{\text{catalog}} = \frac{4\pi N_{\text{avg}}}{FOV^2}$$

To solve this trade-off, we could introduce a constraint on the resolution of the sensor, which limits the FOV for a given detector array, namely the number of pixels utilized for the image representation. Let us go through an example: roughly speaking, if we have a star sensor with FOV angle equal to 20° and a detector of 1024 pixels, we obtain a resolution of $\varepsilon = \frac{FOV}{n_{px}} \approx 0.02 \frac{^\circ}{px}$. We may obtain subpixels resolution depending on the centroiding algorithm that can achieve accuracy of $\gamma = 0.1\ px$ width. In this case, $\varepsilon_{centroid} = \gamma\varepsilon$. The final accuracy of the pointing direction depends on the number of stars in the FOV [28], namely $N = 5$ in this example, as:

$$\text{accuracy} = \frac{\varepsilon}{\sqrt{(N)}} = \frac{FOV}{n_{px}\sqrt{(N)}} \approx 30 \text{ arcsec}$$

Table 6.9 Star sensors strengths and weaknesses.

Strengths	Weaknesses
Extremely accurate	Mass and volume
Provide complete attitude information	Complexity and high cost
Wide applicability across the Solar System	Must avoid luminous objects (e.g., the Sun)
	Requires time to acquire the initial attitude state from a lost-in-space condition

Or, with centroiding accuracy $\gamma = 0.1\ px$:

$$\text{accuracy} = \frac{\varepsilon_{\text{centroid}}}{\sqrt{(N)}} \approx 3 \text{ arcsec}$$

Table 6.9 summarizes the strengths and weaknesses of star sensors.

Performance comparison

Table 6.10 summarizes the typical orders of magnitude of the performance for each presented attitude sensor. Obviously, this is only a rough guideline. A GNC engineer should go through the specifications and the datasheets of each component to acquire precise indications on the actual performance data.



Inertial sensors

Inertial sensors are used to transduce inertial forces into electrical signals, to measure the specific forces — i.e., accelerations — and the angular rates experienced by them and, consequently, by the bodies on which they are installed. Inertial sensors sensing linear acceleration are referred to as accelerometers, while the angular motion is sensed by sensors denoted as gyroscopes. A single accelerometer, or gyroscope, is capable to measure the associated quantity along one axis. Thus, a set of three inertial sensors along orthogonal directions is needed to measure the whole 3D quantity, and it is commonly defined as Inertial Reference Unit (IMU). When triaxial accelerations and angular velocities are provided by a single component, this is referred to as IMU, which is a six-axis sensor containing three orthogonal

Table 6.10 Typical order of magnitude of the performance for each presented attitude sensor.

Sensor	Range of performance
Magnetometer	0.5−5°
Sun sensors	0.001−1°
Horizon sensors	0.1°−0.5°
Star sensors	1 arcsec−1 arcmin

accelerometers and three orthogonal gyroscopes. Sometimes, IMUs can be also equipped with three orthogonal magnetometers (cfr. Section Magnetometers), and, in this case, they are denoted as nine-axis IMU.

Two basic typologies of inertial sensors exist: the "gimballed" and the "strapdown." The former was the first original application of inertial sensors technology, which uses stable platform techniques, where the inertial sensors are mounted on a stable platform and are mechanically isolated from the motion of the vehicle. In this case, triads of accelerometers and gyroscopes are suspended by a system of three gimbals keeping the platform in a fixed orientation with respect to the inertial space. This can be achieved by exploiting conservation laws or with a gyro-stabilized mechanized system. In mechanized gimballed system, the stabilization is guaranteed by three integrating gyros, whose output is proportional to the rotation angles to which the sensors are subjected. Their measurements are used in a feedback cycle commanding null-seeking servomotors, which drive each gimbal to maintain a constant orientation in inertial space. Gimballed inertial sensors directly measure inertial accelerations, and, thus, the displacement from the initial position is computed integrating twice with respect to time. Analogously, the body rotations are immediately retrieved from the gimbal's angles with respect to the platform supports. Despite the very high accuracy that is possible to achieve with this typology of inertial sensors, their technological complexity ties down their usage to systems requiring extremely precise measurements for prolonged time, such as ships and submarines. Purely gimbal sensors are practically unrealistic due to the requirement to have almost null friction in the bearings, while mechanized ones have a considerable impact on mass, volume, power, costs, and maintenance of the system. Moreover, gimballed systems are affected by the gimbal lock, which happens

when the axis of two of the gimbals are driven in the same direction, losing one degree of freedom on the 3D isolation capability (also experienced by the Apollo 11 Moon mission [29]). This problem may be overcome by using a fourth gimbal or by performing dedicated reset rotations. Both solutions further increase the complexity of gimballed inertial sensors.

For all these reasons, from the early 1970s [30,31], the inertial sensors technology was pushed toward alternative, simpler, gimballess inertial sensors. The basic idea is to rigidly attach, or "strap down", gyroscopes and accelerometers directly to the system body, using the gyroscope information to compute the current attitude of the body and, then, generate a coordinate transformation that operates on the accelerometer data to give acceleration resolved into inertial axes. In effect, this results in a "mathematical gimbal set," replacing the mechanical one. The theoretical aspects behind strapdown inertial sensors are two: the measurement of acceleration in body-fixed coordinates and the determination of the orientation of the body-fixed reference frame with respect to inertial one from gyro information. Strapdown accelerometers measure real inertial acceleration, but they resolve it in body-fixed axes. Thus, apparent accelerations arise interpreting these components in the noninertial reference frame. In general, strapdown systems demand for increased computing capabilities, since the attitude transformation computations have to be carried out at a very high frequency. The lack of the inertial platform let the body-mounted instruments experience both the angular and linear motions of the system and most of its vibrations. Moreover, these instruments are subject to the entire maneuver envelope of the spacecraft, and not only to the residual drift rotations of the gyro-stabilized platform. For these reasons, they shall be capable to detect much higher rates of turn, while maintaining good drift accuracy values. Consequently, very broad and linear dynamics range is mandatory for this kind of inertial sensors. Despite these minor drawbacks, strapdown inertial sensors are now widely used in spacecraft applications, and with the current on-board computing capacities, they are a fully mature technology, with performance almost comparable to the gimballed systems.

Gyroscopes measure the changes in vehicle attitude or its angular rate with respect to an inertial reference frame. Accelerometers, however, are not capable to distinguish among the total acceleration of the vehicle with respect to inertial space, and that caused by the presence of a gravitational field. The output of these sensors is the difference between the true

acceleration in space and the acceleration due to gravity. Namely, the nongravitational force per unit mass exerted on the instrument:

$$\mathbf{a}^b = {}_b\mathbf{A}_i \left(\ddot{\mathbf{r}} - \mathbf{g}(\mathbf{r}) \right) \tag{6.2}$$

where the superscript b indicates that it is measured in the body reference frame, $\mathbf{r}$ is the inertial position vector of the accelerometer, $\mathbf{g}(\mathbf{r})$ is the position-dependent gravitational acceleration, and ${}_b\mathbf{A}_i$ is the direction cosine matrix rotating a vector from the inertial to the body reference frame. Hence, in free-fall orbiting applications, accelerating due to of the gravity of celestial bodies only, an ideal accelerometer placed at the center of gravity of the spacecraft would measure zero. Then, to compute the total acceleration acting on the body, which is needed to propagate its motion, a gravitational field model shall be used to correct accelerometer measurements with gravity accelerations. The navigation functions, discussed in Chapter 9 — Navigation, exploiting strapdown inertial sensors measurements shall combine body rotation estimation, reference frame transformation, gravitational acceleration evaluation, translational and rotational coupling correction, and dynamics integration, as in Fig. 6.25.

Accelerometer's technology is well established since many decades, and it is based on mechanical sensors or solid-state ones. Despite the construction principle of these sensors, they are usually based on Newton's second law of motion to measure the force acting on a small mass, known as a proof or seismic mass, which is contained within the vehicle, and it is typically connected via a spring or an oscillating arm to the case of the instrument. When

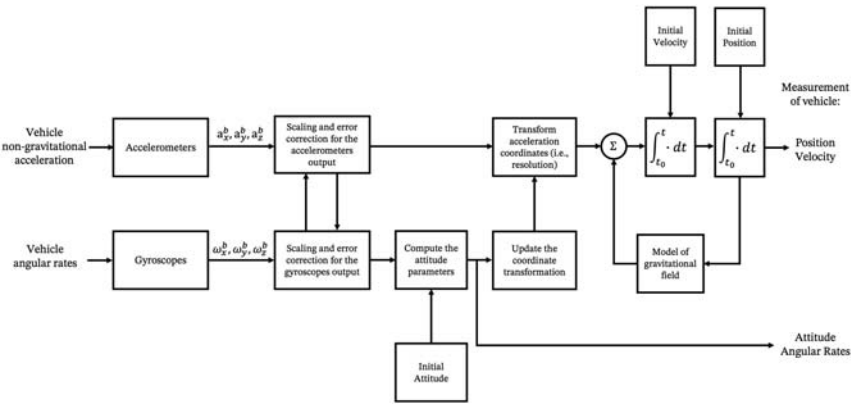


Figure 6.25 Functional components of a navigation system based on inertial sensors.

the instrument is subjected to an acceleration along its sensitive axis, the proof mass tends to respond to the change in movement according to its own inertia. A transducer is designed to sense the relative movements between the proof mass and the sensor case, converting them in calibrated electric signals proportional to the nongravitational forces acting on the body. As already said, an accelerometer freely orbiting around a celestial body would output a zero measurement, since both the case and the proof mass experience a common influence of the surrounding gravitational field. In this case, the acceleration of the instrument with respect to an inertial reference frame is $\mathbf{a} = \ddot{\mathbf{r}} = \mathbf{g}$, and the specific force is zero in accordance with Eq. (6.2). The principles of operation of accelerometers depend on the construction technology, which can be based on: force-feedback pendulum, optical fiber pendulum, vibratory quartz crystal technology [32], acoustic wave resonator on piezoelectric cantilever beam [33], silicon-based pendulum [34], interferometry [35], optical waveguides and Bragg gratings [36]. In recent years, MEMS technology began to be popular for inertial sensors. This construction principle will be discussed in the following, together with gyroscope MEMS technology, but high-accuracy accelerometers will continue to incorporate mechanical sensors, with some use of resonant devices [32].

Gyroscope's technology is typically divided in two broad classes: rate gyros that read angular rates and rate-integrating gyros that measure integrated rates or angular displacements. The most basic gyroscope construction principle is based on the angular momentum conservation principle of a wheel, or rotor, spinning at high speed. In this case, the instrument contains a spinning mass, whose angular momentum vector, coincident with the axis of spin of the rotor, remains fixed in the inertial reference frame. When the sensor is subjected to a rotation along its sensitive axis, a precession displacement between the instrument case and the spinning axis is sensed by a transducer converting it in a calibrated electric signal. In alternative, the precession principle can be also applied to accurately measure the rotation rates, by applying a torque to maintain the angular momentum of the proof mass fixed in the body frame. Then, the measurement of this torque provides measurements of the angular velocity of the instrument, according to:

$$\mathbf{t}^b = {}_b\boldsymbol{\omega}_i^b \times \mathbf{h}^b, \tag{6.3}$$

where $\mathbf{h}^b$ is the gyro's angular momentum, ${}_b\boldsymbol{\omega}_i^b$ is the angular rate of the body with respect to the inertial reference frame, and $\mathbf{t}^b$ is the output torque

measured in body frame. Single-axis gyros sense only one direction of the torque, and the other reaction torque is provided by mechanical constraints on the orthogonal axis. Very accurate spinning mass gyros are single-axis floated devices, where the rotor is contained in a case that is immersed in a buoyancy fluid to reduce the load on the gimbal bearings [32]. Two-axis gyros are called dry tuned-rotor gyros (DTGs) because they are not floated. They have two input axes which are mutually orthogonal, and which lie in a plane which is perpendicular to the spin axis of the gyroscope. The rotor deflection is sensed on both axes, and it is controlled electromagnetically according to Eq. (6.3). DTGs have come very close to the performance of single-axis floated gyros, but never achieved performance equivalent to the best floated gyros [37]. However, they are simpler, rugged, and less prone to failures.

Other typologies of gyroscopes are based on direct rate sensing on viscous coupling between fluids and masses [32]; vibration motion deflection due to Coriolis apparent forces, where a vibration element (e.g., a string, a hollow cylinder, a rod, a tuning fork, a hemispherical dome) is present to sense rotation-induced deflections; optical interferometry and Sagnac effect. Coriolis vibratory gyros, especially in the tuning fork version, are fundamental for the size reduction of gyroscopes, and the realization of silicon solid-state or MEMS sensors. They will be discussed in the following. Optical gyroscopes allow extremely high performance and accuracy since they measure optical length differences within a circular waveguide. In fact, when light travels in opposite directions (clockwise and anticlockwise) around an enclosed ring, differences arise in the apparent optical length of the two paths when the ring is rotated about an axis perpendicular to the plane containing the ring itself. The path differences can be practically detected as a phase shift, but, even in this case, the angular differences are extremely small [23]. To overcome this practical problem, interferometric fiber optic gyros and ring laser gyros have been ideated. The former use many turns of optical fiber to magnify the phase shift difference, the latter replace the phase measurement by a frequency difference measurement between two counterpropagating resonant laser beams. Despite their potential, optical sensors are not widespread in spacecraft applications since they require maintenance to overcome optical degradation, they are power consuming, they lose detection capability for very slow rotation rates, and they have a complex implementation to achieve good performance.

Recent technology developments allowed simpler, smaller, lighter, lower power consuming, and lower cost inertial sensors. These features

are all positive in spacecraft applications, but they become fundamentals for small satellites and CubeSats (cfr. section CubeSats in Chapter 15 — Modern Spacecraft GNC), and they are possible, thanks to MEMS sensors [38,39]. MEMS sensors are packaged similarly to other integrated circuits, with a single part containing inertial sensors for multiple axes. They are produced with chemical etching and batch processing techniques, and they were recently qualified for operations in the space radiation environment. The major drawbacks of MEMS sensors are related with their short lifetime and reduced performance, compared to other inertial sensors. In fact, size reduction is commonly related to a decrease in sensitivity and an increase in noise. MEMS sensors are made of silicon wafers or piezoelectric quartz layers, and they are based on the same physical principles of the classical inertial sensors. MEMS gyroscopes exploit the Coriolis acceleration effect on vibrating proof masses to detect angular rotation, while MEMS accelerometers exploit miniaturized pendulum displacements or microresonators vibration frequency changes to detect linear acceleration. An array of MEMS sensors may be integrated into a single chip to provide multiple independent measurements of inertial motion. Moreover, single-chip implementations of six-axis, or nine-axis, IMUs have become common, guaranteeing a vast reduction in volume, negligible power consumption, and the ability to carry out a complete characterization of the unit in a single operation.

Typical error sources

Every inertial sensor is subject to errors which limit their measurement accuracy. The applications of these instruments in spacecraft shall consider their common error sources in order to minimize their impact on the GNC functions. Moreover, a proper GNC design shall rely on a proper sensor modeling that considers all the relevant errors affecting the measurements. The error sources of inertial sensors can be categorized in four classes:

- Deterministic (or constant) errors, which are repeatable. They can be determined and corrected by calibration.
- Temperature-induced errors, which can be determined and corrected by thermal calibration.
- Switch-on to switch-on errors, which are stochastic processes but stays constant for any single run. They can be corrected with on-board calibration estimation techniques, discussed in Chapter 14 — Applicative GNC Cases and Examples.

- In-run stochastic variations errors, which should be modeled as stochastic processes varying throughout the measurement run. In-run stochastic variations with slow dynamics can be corrected with on-board calibration estimation techniques. Stochastic errors with a fast dynamics (i.e., random noise) cannot be corrected.

Typical error sources depend on the specific type of sensor. However, the major ones are in common to all of them.

Bias errors are related to a nonnull sensor output, which is present even in the absence of an applied input. They are typically expressed in units of degrees per hour (°/h) for gyroscopes, and milli-g or micro-g (mg, μg) for accelerometers.

Bias errors in both gyroscopes and accelerometers can be further broken down into more error components:

- *Constant bias:* fixed bias term in every measurement of the sensor. It can be corrected by calibration of the sensor.
- *Bias repeatability* (switch-on to switch on bias): bias term varying at every sensor powerup. It remains constant until next switch-on. It can be corrected by in-run calibration with bis estimation techniques. A high bias repeatability guarantees a faster convergence of the estimation filters between distinct runs of the sensor.
- *Temperature-dependent bias:* bias term depending on the operating temperature of the sensor within the specified temperature range. It can be corrected by implementing a thermal model of the sensor and taking into account temperature measurements of the instrument.

Mechanical gyroscopes are also affected by *acceleration-dependent bias* (i.e., g-dependent bias), which is proportional to the magnitude of the applied acceleration. Such errors arise because of mass unbalance and asymmetry, due to imperfections caused by the fabrication process. They are expressed by means of coefficients having units of degrees per hour per g (°/h/g).

Bias instability errors are related to an in-run bias drift with stochastic evolution, which can be characterized in terms of variance and process time. Bias instability has an impact on the long-term integration of sensor's outputs. It is expressed in units of degrees per hour (°/h) for gyroscopes, and micro-g (μg) for accelerometers. It is defined with the AVAR method, which is introduced in the following dedicated section. Bias instability determines continuous fluctuations in the bias offset, which can be estimated on-board to improve motion propagation performance.

Random walk errors are pure stochastic processes associated to the characteristic of the intrinsic noise of the sensor. Integration of the random walk

noises in the measurements leads to a random walk in the final solution. They can be estimated by the AVAR method, which is introduced in (cfr. Section AVAR). Noise errors are defined according to the noise variance trend with respect to the averaging time.

- *Angle random walk (ARW)/Velocity random walk (VRW) errors* are high-frequency noises, and they can be observed as the short-term variations in the output. ARW is defined for gyroscopes, while VRW is defined for accelerometers. Integrating the sensor outputs, these terms cause random errors in angle/velocity estimation with a distribution that is proportional to the square root of the elapsed time. Thus, these errors will increase the longer integration and provide a fundamental limitation, together with bias instability, to any angle/velocity measurement that relies only on inertial sensors. They are expressed in units of degrees/meters per second per square root of hour ($^\circ/\sqrt{\text{h}}$, $\text{m/s}/\sqrt{\text{h}}$ or $\mu\text{g}/\sqrt{\text{Hz}}$). Note that, VRW can be also expressed in units of micro-*g* per square root of Hertz.
- *Rate random walk (RRW) errors* are long-term changes to the bias offset, which are randomly distributed over very long time. The time scale over which these changes occur can be defined by the RRW and allow for on-board recalibration of the sensor.

Random walk errors are classified in terms of characteristic time, and they are further differentiated from those associated with bias instability because of the different temporal properties, as analyzed in the power spectral density of the sensor stochastic processes.

Scale factor errors affect the ratio relating the change in the output signal to a change in the input to be measured. It is the measure of the gradient of the best straight line that can be fitted by the method of least squares to the expected sensor output signal, against input inertial signal over the full dynamic range at ambient temperature. Scale factor errors are commonly expressed as a ratio of output error to input quantity, in parts per million (ppm) or percentage. Additional errors arise from the nonlinearity of the scale factor with respect to the least squares linear fitting, and from its asymmetry with respect to opposite direction inputs.

Scale factor ratios are typically not constant and may fluctuate during in-run operations. *Scale factor instability* is an additional source of errors.

Cross-coupling errors determine spurious outputs resulting from sensor sensitivity to accelerations or angular rates about axes orthogonal to the input axis. Such errors arise through nonorthogonality of the sensor axes.

They are expressed as parts per mission (ppm) of the applied input. They can be corrected by calibration being repeatable deterministic errors.

Misalignment errors are related to a mechanical mounting error of the sensor triad with respect to the nominal body frame. The sensor outputs have couplings with respect to different body axes inputs. They are expressed in units of milliradians (mrad), and they are difficult to be distinguished with respect to cross-coupling errors. They can be corrected by calibration being repeatable deterministic errors.

Other than the previous sources of error, inertial sensors are characterized by other limitations, such as:

Input range limits associated to the maximum input values that can be meaningfully measured.

Output quantization associated to the mapping of a continuous signal into a discrete one and dependent on the analog to digital conversion resolution.

Latency associated to the sensor processing time, which determines a nonzero elapsed time between the input sensing and its output generation.

Moreover, inertial sensors are also affected by extrinsic errors due to the estimation processes elaborating inertial sensors outputs, as seen in Fig. 6.25. For instance, an imprecise model of the gravitation field may lead to motion propagation errors. Similarly, numerical computation errors may also affect the precision of propagated position, velocity, or attitude vectors. However, these latter errors are associated to the navigation functions and not directly to the inertial sensor itself. Fig. 6.26 summarizes the main sources of errors described in this section.

Inertial sensors performances

Inertial sensors are classified into various performance grades according to their typical accuracy levels, which are divided in four main performance categories:

- Consumer grade.
- Industrial grade.
- Tactical grade.
- Navigation grade.

These categories are mainly subdivided according to the in-run bias stability of the sensor, as the in-run bias stability plays such a large role in determining inertial navigation performance. In fact, a low value of the bias drift enables the application of inertial sensors in high-end market segments. The performance grades, associated to the available sensor technologies, are

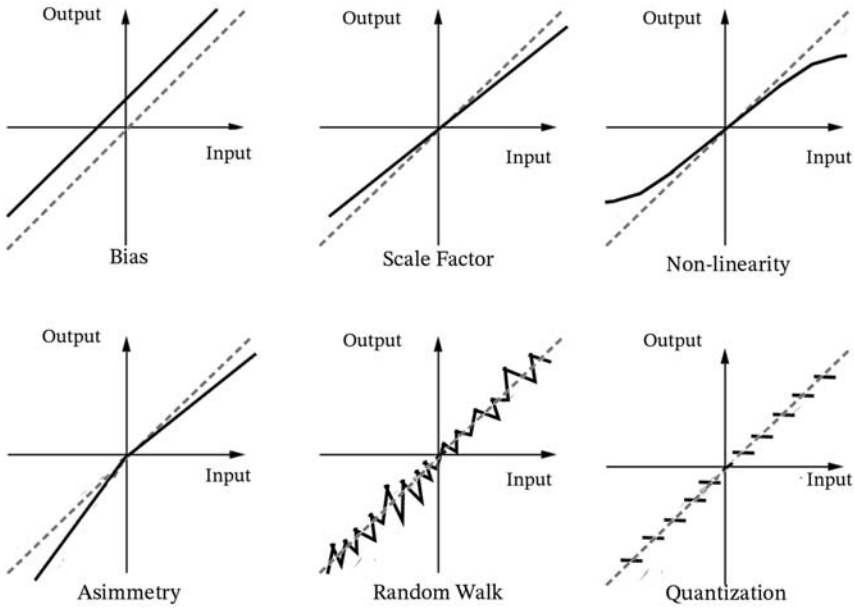


Figure 6.26 Inertial sensor errors.

reported in Figs. 6.27 and 6.28 [40,41]. Given the different performance of the sensors, the order of magnitude for position and angle accuracy estimation, after 1h of motion propagation, is reported in Table 6.11.

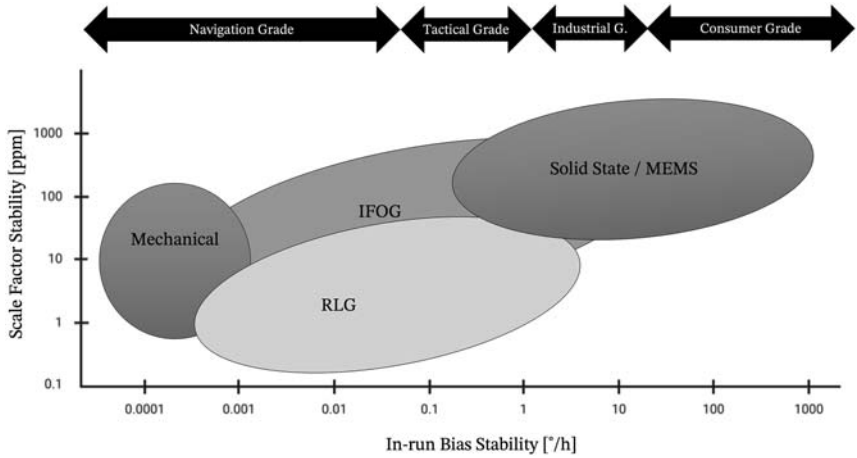


Figure 6.27 Gyroscope technologies and performance grades.

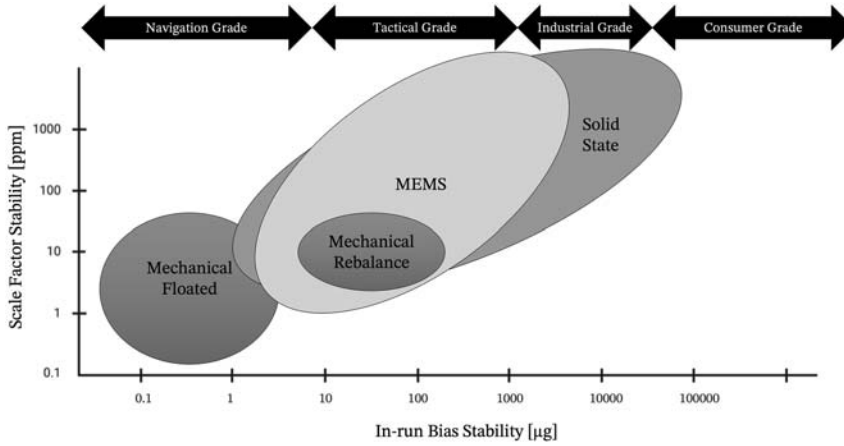


Figure 6.28 Accelerometer technologies and performance grades.

Table 6.11 Sensor grades and navigation performance after 1h of propagation.

Sensor grade	Angle accuracy (°)	Position accuracy (km)
Consumer grade	>50	>1000
Industrial	5−50	100−1000
Tactical	0.1−5	1−100
Navigation	<0.1	<1

Allan variance and statistical error representation

Inertial sensors are affected by both repeatable deterministic errors and stochastic random errors. The former can be eliminated or reduced, thanks to standard sensor calibration techniques, while the latter are more difficult to be compensated, unless complex statistical estimation techniques are used. However, in general, it is almost impossible to completely remove their impact on the motion propagation performance.

In-run stochastic variations are due to temperature, aging, or mechanical stress on the system. Furthermore, electronic noise, interactions with the external environment, and transduction interferences are additional issues generating random errors on inertial sensors output. Thus, given this broad range of physical phenomena, stochastic error sources have different characteristic times, filling a complete frequency spectrum. In fact, they appear on the output as a noise or a slow change of parameters in time. The contribution of these errors at a specific time cannot be precisely predicted, but the analysis of these stochastic processes in inertial sensors is possible. According

to the international IEEE standard 952−1997 [42], two analysis methods are available, but the AVAR method is the most widely used. AVAR is the method of analysis of stochastic processes in a time domain, and it describes the intrinsic variance of a signal as a function of averaging time. It can evaluate the real performance of inertial sensors in relation to the effects of bias instability, ARW and RRW. Note that, often the AVAR term is also used to refer to its square root.

To compute the AVAR, we have to follow the following steps [40]:

1. *Data collection:* sample the output data, for a steady-state input condition, from the sensor at interval T_s to obtain a long sequence of consecutive N-point data.
2. *Data clustering:* divide the N-point data into K clusters, with each cluster having the length of M-point data.
3. *Data averaging:* average the data in each K-th cluster.

$$\bar{a}_K(M) = \frac{1}{M} \sum_{i=1}^{M} a_K(i)$$

4. *Calculate variance:* take the difference in average between successive clusters. There must be enough data for at least nine bins, otherwise the results obtained begin to lose their significance. The AVAR is defined as the mean value of the square of the difference of adjacent time averages from a time series as a function of averaging time, multiple of the sample time, $\tau = mT_s$, and it is expressed as:

$$\sigma^2_{\text{Allan}}(\tau) = \frac{1}{2(K-1)} \sum_{K=1}^{K-1} \left[\bar{a}_{K+1}(M) - \bar{a}_K(M)\right]^2$$

A typical Allan plot for inertial sensors is shown in Fig. 6.29, in which the values of the root AVAR are plotted against the averaging interval, τ, in log-log form. Different types of random processes cause slopes with different gradients to appear on the plot, and they also are correlated with different averaging times, and therefore correspond to different locations on the Allan plot. The basic noise terms are ARW, RRW, bias instability, quantization noise, and drift rate ramp. In addition, the sinusoidal noise and exponentially correlated noise can also be identified on the plot.

As shown in Fig. 6.29, the electronic quantization white noise is characterized on the AVAR plot as a slope with gradient equal to −1. The magnitude of this noise can be read off the slope line at $\tau = \sqrt{3}$. This noise is strictly due to the digital nature of the sensors. The random walk

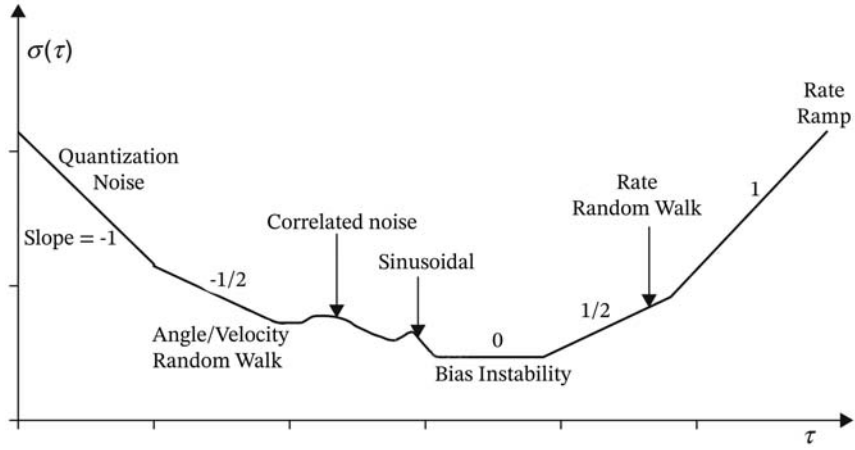


Figure 6.29 Example Allan variance analysis results adapted from [42].

measurement for sensor noise, such as the ARW for gyros and VRW for accelerometers, can be obtained directly by reading the slope line at $\tau = 1$. These noise terms are all characterized by a white noise spectrum on the output. Bias instability is due to low frequency bias fluctuations, which are characterized by $1/f$ noise or flicker noise spectrum, where f is the cutoff frequency of the sensor output. Bias instability appears on the plot as a plateau around the minimum. The flat region of the plot can be examined to estimate the limit of the bias instability, as well as the cutoff frequency of the underlying flicker noise. The RRW indicates a random walk with a very long correlation time. It is represented on the AVAR plot by a slope of $+1/2$. The magnitude of this noise, K, can be read off the slope line at $\tau = 3$. For long, but finite time intervals, this is more of a deterministic error rather than a random noise. Its presence in the data may indicate a very slow monotonic change in the sensor properties, and its magnitude can be obtained from the slope line at $\tau = \sqrt{2}$.

If it can be assumed that the existing random processes are all statistically independent, then it can be shown that the Allan variance at any given t is the sum of Allan variances due to the individual random processes at the same τ:

$$\sigma^2_{Allan}(\tau) = \sigma^2_{ARW}(\tau) + \sigma^2_{Quant}(\tau) + \sigma^2_{BiasInst}(\tau) + \sigma^2_{RRW}(\tau) + \cdots$$

Thus, estimating the amplitude of a given random noise in any region of the AVAR requires a knowledge of the amplitudes of the other random

noises in the same region. Fig. 6.30 reports the individual AVAR contributions of the main random processes discussed above.

Note that, the computation of AVAR needs a finite number of clusters that can be generated from the raw data measurements of the sensors. Depending on the size of these clusters, the AVAR can identify any noise term that is affecting the data sensor. It is important to mention that the estimation accuracy of the AVAR for a given t depends on the number of independent clusters within the data set. The bigger the number of independent clusters, the better the estimation accuracy.

All the stochastic errors contribute to accumulate errors, with an increasing standard deviation, along the motion propagation. If a continuous bias estimation were possible, the propagation errors would theoretically approach zero, in absence of other noncalibrated errors. However, this is not feasible, and finite time periodic calibrations are the best viable solution. The frequency of these in-run on-board calibrations can be selected from the AVAR analyses. In fact, calibrating at least before the bias stability characteristic time makes ARW dominant and drift negligible. Given the time interval at which we can effectively recalibrate, we find out from the AVAR graph the statistical estimate of the propagation error. For example,

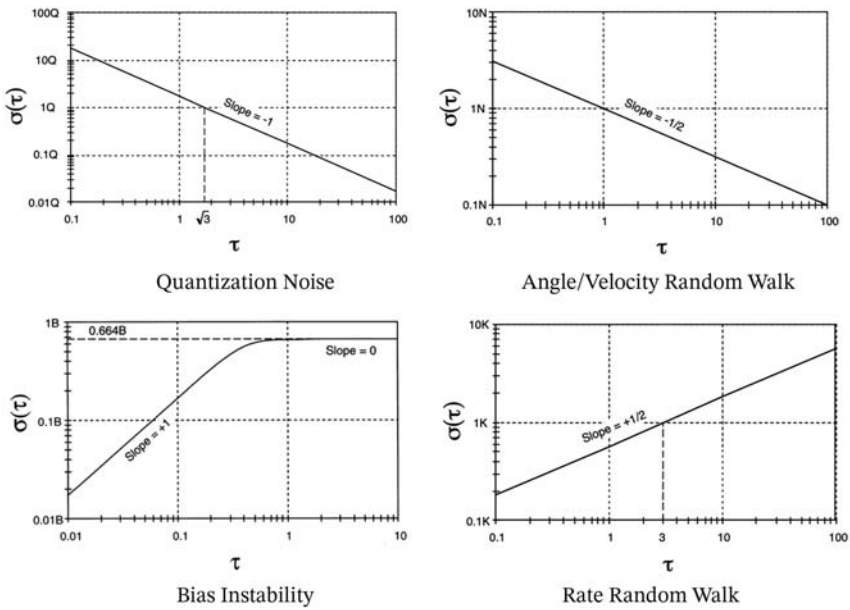


Figure 6.30 Individual Allan variance contributions adapted from Ref. [42].

assuming the root of the AVAR of a tactical grade MEMS gyroscope at $\tau = 2700$s being $\sigma_{\text{Allan}} = 1.3\,^{\circ}/\text{h} = 361\,\mu^{\circ}/\text{s}$, and propagating the motion along a 45 min $= 2700$ s window (e.g., an LEO eclipse period), the statistical propagation error would be $\varepsilon_{\theta} = 0.9747\,\text{deg}_{RMS}$.

As said, two main parameters represent the quality of inertial sensors. They are the ARW and the bias for gyroscopes, and the VRW and bias for accelerometers. Noting that bias and scale-factor drifts are much larger than noise contributions, Table 6.12 reports random walk ad bias values for typical inertial sensors.

Remind that inertial sensor units are note standardized, and different conversion factors are very useful to be known. Working with the square root of time, converting between hours and seconds is a factor of 60 instead of 3600: $60\sqrt{\text{s}} = \sqrt{\text{h}}$. Similarly, accelerations can be expressed in μg or in the SI m/s/s, and $1\mu\text{g} \approx 10^{-5}\text{m/s/s}$. Thus, for example, a VRW of X μg/$\sqrt{\text{Hz}} \approx X\ 6 \times 10^{-4}\text{m/s}/\sqrt{\text{h}}$.

Gyroscope model

A general gyroscope model including all the relevant sensor's errors is:

$$\boldsymbol{\omega}(t) = (\mathbf{I}_3 + \mathbf{SF})\mathbf{OR}\ \boldsymbol{\omega}_{true}(t) + \mathbf{b}(\text{t}) + \boldsymbol{\eta}(t).$$

An example of this model implementation in MATLAB/Simulink is shown in Fig. 6.31. A similar structure can be extended of other sensor models, such as accelerometers or magnetometers. The implementation of the individual error sources would be the same in all the sensor models affected by the same errors. Note that bias and noise terms can also be

Table 6.12 Typical random walk noise and bias values for inertial sensors.

Sensor grade	Gyro bias (°/h)	ARW noise (°/$\sqrt{\text{h}}$)	Acc bias (μg)	VRW noise (μg/$\sqrt{\text{Hz}}$)
Consumer grade	>50	>0.5	>50,000	>100
Industrial	5−50	0.1−0.5	1000−50,000	10−100
Tactical	0.1−5	0.05−0.1	10−1000	1−10
Navigation	<0.1	<0.05	<10	<1

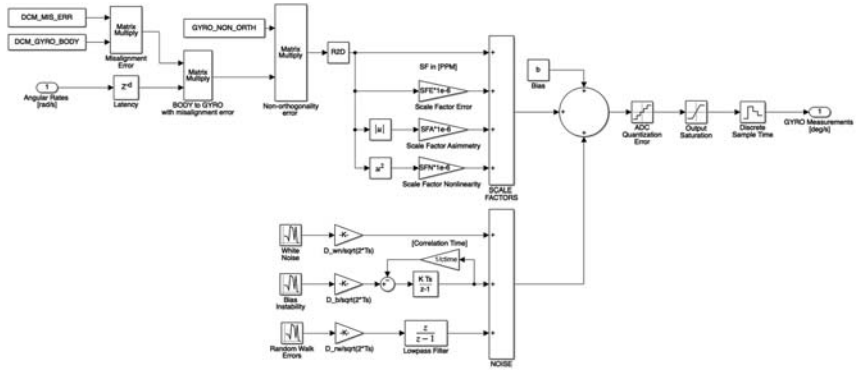


Figure 6.31 Gyroscope model.

multiplied by the scale factors, misalignment, and nonorthogonality errors (i.e., $(\mathbf{I}_3 + \mathbf{SF})\mathbf{OR}$). It is just a matter of their definition and implementation. In Chapter 14—Applicative GNC Cases and Examples, the other formulation is used to simplify the sensor calibration equations.

Electro-optical sensors

For unmanned missions, spacecraft GNC requires a level of autonomy that can only be provided, thanks to dedicated on-board GNC including electro-optical sensor suites. In fact, the measurements taken by this kind of sensors enable autonomous navigation, providing on-board data that are fundamental to have the spacecraft estimating its state.

To achieve the high level of autonomy and reliability required in today's space missions, navigation making use of high-accuracy measurements, to be processed on-board at high frequency, is mandatory in critical operations. The mission concept and the navigation technique will drive the selection of different sensor suites.

In general, the term electro-optical indicates all the instruments capable of collecting reflected or emitted radiations in a certain range of the electromagnetic spectrum. In modern spacecraft applications, almost exclusively visible (0.37—0.75 μm) and infrared (0.75—1000 μm) bands are used, and the most common sensors are cameras (monocular and stereo) or LIDAR systems.

Cameras

Cameras are based on passive electro-optical sensors that collect incoming light waves (visible or infrared) and traduce them into signals which allow to produce an image. The two main types of electro-optical sensors employed by modern cameras are the charge-coupled device (CCD) and the active-pixel sensor (complementary metal oxide semiconductor (CMOS)). Without entering into technical details, both CCD and CMOS are composed by arrays of silicon photosites, also known as pixels, that convert collected photons into electrical signal. In a CCD sensor, each photosite is an analog device and the charge is transported across the chip and read at one corner of the array. An AD converter is used to convert the analog signal into a digital value. On the other hand, CMOS sensors rely on one or more transistors at each photosite that amplify and move the charge using more traditional wires. The main difference in terms of performance can be summarized here:

- CCD sensors are superior to CMOS for what concerns high-quality, low-noise images.
- Light sensitivity of a CMOS chip tends to be lower, as many of the photons hit the transistors instead of the photosite.
- CCD sensors require much higher power than an equivalent CMOS sensor (up to 100 times more).

Another important classification while referring to imaging sensors is the type of shutter adopted. The shutter is the device (mechanical or electric) allowing light to hit the sensor for a given amount of time (i.e., exposure time). Two main types of shutters can be identified: rolling and global. Rolling shutters expose sensor rows sequentially from top to bottom, while for global shutters, all the pixels of the sensors are exposed at the same time. For fast dynamics navigation scenario, it is generally preferred global (snapshot) shutter mode as rolling mode introduces many artifacts shifting in the image (smearing effect), and global shutter avoids preprocessing compensation. When the camera is moving very fast or spinning with respect to the target, with a roller shutter, the image is distorted because different lines can capture different times of the scene, generating a "moving picture." One classical example of this undesired phenomenon is obtained while imaging the rotor of a helicopter. The rotor blades would appear to be unnaturally swept back due to the rolling shutter.

The optical lens (or a group of lenses) will complete the camera assembly. A lens is characterized by a set of two main parameters:

- Focal length: it determines the magnification of the image projection onto the image plane. It is also inversely proportional to the camera FOV, i.e., longer focal length will correspond to narrower FOV.
- Aperture: it determines the amount of light hitting the camera sensor.

Finally, a very important figure of merit, especially for space applications is the signal-to-noise ratio (SNR). The SNR represents a quantitative value describing the quality of a measurement. As the name suggests, SNR is obtained as a ratio between the measured signal and the overall measured noise. High values of noise imply very good measurement of the target object, i.e., the incoming signal is very strong with respect to the overall noise. On the contrary, low SNR can produce invalid measurements and consequent target undetectability. This is the case, for example, of very far objects having a very low signal measured by the sensor. In fact, the incoming signal directly depends on the photon flux hitting the sensor (therefore on the integration time, sensor quantum efficiency, and lens optical transmittance). On the other hand, the sources of noise are almost exclusively dependent on the physical characteristics of the sensor. The main noise sources are summarized in Table 6.13.

Table 6.13 Typical noise sources.

Noise	Description	Unit
Read noise	Inherent to the process of converting charge carriers into a voltage signal for quantification, and analog-to-digital conversion.	e^-
Dark current	Represents the level of thermally generated electrons, directly dependent on exposure time and temperature.	$\frac{e^-}{s}$
Signal shot noise	Inherent natural variation of the incident photon flux.	$\frac{e^-}{s}$
Sky/background noise	Overall optical noise of the system, composed by the electronic noise in the sensor, and undesired stray light.	$\frac{e^-}{s}$

Applicability

Camera systems have the advantage of being inexpensive, passive, compact, and power efficient. Furthermore, they provide images at high frame rates that are readily interpretable by human operators for improved situational awareness. Cameras are typically usable at ranges of a few centimeters up to around 100 m, with the exact range being determined by the interplay among the camera imaging resolution, field of view size, physical dimensions of visual targets, and the employed features (i.e., centroid, shape, reference points). On the downside, their image output should be processed by complex algorithms for extracting cues guiding the estimation of spacecraft state. These algorithms can have low robustness and are often computationally demanding. Camera-based approaches should also deal with foreground segmentation in order to isolate the relevant sensory input from its potentially confusing background. Furthermore, to operate in low light conditions, camera sensors have to depend on light emitted from external sources. Thermal cameras have also been proposed to avoid illumination problems or exploited as aide to the main optical system even though they offer much lower accuracy [43]. Approaches using a single camera are referred to as monocular. However, the low cost of camera units permits pose estimation using two or even more such sensors, giving rise to approaches referred to as binocular or multiocular, respectively. A list example of vision-based navigation scenarios may include planetary approach application, satellite servicing, active debris removal, Descent and Landing scenario (high or low dynamics), general rendezvous, and docking application or rovers' navigation (localization and mapping). In most of the missions, the navigation cameras will provide a gray-scale image in the visible wavelength. Depending on the FOV of the camera, we will refer to Narrow-Angle Cameras (NACs) or Wide-Angle Cameras (WACs). We refer to NAC when the optics design provides an FOV in the range of $1-10°$. On some occasions, but is not very common, a Medium-Angle Camera (MAC) may be defined, with FOV in between 10 and 30°. Finally, a WAC usually has an FOV larger than 20°. NAC will be used when the target is very far from the observer. Having a smaller FOV implies to have a larger target apparent dimension on the camera sensor. This is useful at far range, but it can become problematic at close range when the target may occupy all the FOV dimension or high accuracy pointing is necessary to keep it in view. WAC will be used for medium to close distances. The closer the target is, the wider the FOV may be necessary to satisfy target object or surface visibility inside the FOV. A larger FOV will also be considered

in certain scenarios where the target has fast dynamics and crosses the detector. In this case, a larger FOV allows to capture the streak of the target in one or more consecutive images granting navigation data extraction. We shall also note that wide FOV, as for instance 50° or 70° introduces high image distortion. The narrower the field of view, the larger the baffle design is needed. Baffle cone will protect the optics and image from light sources outside of the FOV, generating undesired stray lights. A WAC or NAC camera will normally offer not only images in visible wavelength but also in many cases, it also covers near-infrared. When the mission accomplishment will require operations in eclipse or poor illumination conditions, either active illuminance beam is needed or Thermal Infrared (TIR) cameras. Finally, different studies are also analyzing the utilization of Multispectral or Hyperspectral cameras for navigation [44]. Visible and TIR is the most convenient combination, while hyperspectral cameras may better facilitate a reused concept of instrument to be used both for navigation and for scientific purposes as characterization or different science observations. For rovers' navigation or for close operations where 3D information is required, stereovision cameras are selected. Stereo cameras are composed by two optical heads aligned, temporally synchronized and with certain distance separation granting overlap of the FOV. In certain missions, the images will only be acquired on-board while information extraction process is performed on ground, because of the on-board computing constraints. On-board autonomy is mandatory for critical operations, such as scenarios involving fast dynamics or huge delays and latency in communication that would make ground operations impossible. As a clear example, ground operations are impossible if the spacecraft GNC is dealing with descent and landing operations on a small body, which requires to extract data from images at the slowest frequency of 1 frame per second, and the spacecraft to or from ground communication takes 9 min. The on-board processing in that case is mandatory, under a big level of autonomy, and it requires fast image acquisition by navigation cameras, as well as on-board camera image corrections (preprocessing), before the data extraction. Ground operations may acquire raw images and perform the image preprocessing correction offline to have a more accurate output, while for on-board processing, these corrections should be applied on the fly. The image preprocessing corrections include radiometric and geometric calibration. Radiometric calibration is performed to produce an image that ideally would correspond to a perfect detector while the geometric calibration corrects distortions produced by the finite-aperture optics and motion. Optionally, it is also possible to

validate the images in order to detect (and discard) "contaminated" images that may degrade the optical navigation performance. In fact, processing only the best shots improve both the navigation accuracy and the robustness. The low-level image correction functionalities should be, for example, in charge, but not limited to, of the defective pixel correction, photoresponse nonuniformity correction, dark signal correction, background noise reduction due to ionizing particle events or others, motion or aberration correction, and the focal length scale factor calibration. The preprocessing functions should be considered in the sensor suite selection. The navigation cameras should also provide certain metadata together with the images, for the proper identification and relevant information used in the navigation processing. These metadata include health status, self-tests, configuration options, thermal measurements, and very important, a time stamp of the image. There are several camera operation parameters that should be configurable for the proper operation, such as integration time, window position and size, start of integration, and acquisition options as binning, cropping, or even image stacking (stacking improves the SNR and help faint objects detection in trade-off configuration with integration time). The resolution in pixels vary and should be traded-off in between the need for large amount of information and the computational effort that would require processing it. Typical pixel resolutions trade-off selection uses images of 2048×2048 or 1024×1024 pixels. Among the most interesting parameter to consider in the camera selection is the pixel size, image area, frame rate, quantum efficiency, noise, modulation transfer function (MTF), environmental endurance, and shutter property.

A navigation camera for space shall provide good SNR (high sensitivity, low noise), fast frame rates, high pixel count, and low power consumption, as well as being cost-effective production. Both the rolling shutter artifacts as well as the global shutter drawbacks can be compensated, but this means adding logic and complexity, translated in cost.

Design

A navigation camera design shall encompass the following steps:

- Image sensors selection. The sensor size has to be selected according to the required accuracy, considering the mission scenario and desired navigation performance.

- SNR analysis and magnitude detectability. Given the target characteristics and the mission scenarios, an evaluation of the object magnitude and of the resulting SNR is necessary to guarantee observability.
- Optical design. It is the process of selecting and assembling optical components to satisfy the lens requirements. This step is necessary when commercial lenses are not available for the desired application.
- Stray light assessment and baffle design. Light sources (e.g., Sun, planets) can generate undesired stray light hitting the sensor, degrading imaging, and consequently, navigation performance. For this reason, the stray light sources should be identified for the given mission profile and a proper baffle should be designed to limit this undesired effect. In fact, baffles are mechanical systems that, through a series of internal reflections, can shield the incoming stray lights.
- Thermal design. Camera working domain must be ensured through proper thermal regulation systems to guarantee expected performance.

Additional mechanical features as pointing structures may not be directly related to the camera but to the assembly in the satellite, either under the responsibility of the satellite bus platform or under the GNC or AOCS responsible.

Table 6.14 summarizes the strength and weakness of cameras:

LIDAR

LIDAR is a type of electro-optical sensor that use beam-steering devices to actively direct a laser beam in particular directions and rely on different physical principles (namely triangulation, phase shift, or time-of-flight) to measure range. An example of the working principle of scanning LIDARs is represented in Fig. 6.32.

Table 6.14 Cameras strengths and weaknesses.

Strengths	Weaknesses
Reliable	Highly sensible to illumination conditions
Affordable	Not robust to the presence of celestial body in the field of view
Low power consumption	
Low hardware complexity	
Can be used for supervised applications	

The primary advantage of laser-based sensors is that they offer highly accurate output, which is often combined with a large operating range. Depending on their operating principle, their operating range is between a few meters (triangulation) to several kilometers (time-of-flight). The operating principle also influences the sensor speed, i.e., the number of point measurements that can be accumulated per second. Besides taking longer to complete a scan, a low scanning speed also implies that measurements can be distorted due to the relative motion between the sensor and the target during the scan. This, however, is overcome by the recently introduced flash (or detector array) LIDARs, which achieve high frame rates by emitting light pulses within their entire field of view, bypassing scanning. The working principle for this kind of LIDARs is depicted in Fig. 6.33.

Laser scanners readily provide 3D measurements which are also insensitive to illumination changes and shadows. Their primary drawbacks are that they are expensive, bulkier, less power efficient, and have a lower resolution and a small field of view. Moreover, they imply higher computational cost to process a bigger amount of data, usually point clouds. These drawbacks are particularly true for flash LIDARs, which require many times more instantaneous power than scanning LIDARs because they need to divide the power across many detectors of the pixel array, but they don't have any moving parts. This division limits practical combinations of their range, FOV size, and resolution. As a general guideline, flash LIDARS are better suited to scenarios involving faster objects within a smaller volume. It is also noted here that a small field of view might in turn constrain a chaser's approach trajectory toward a target spacecraft. A summary of the main advantages and drawbacks is reported in Table 6.15.

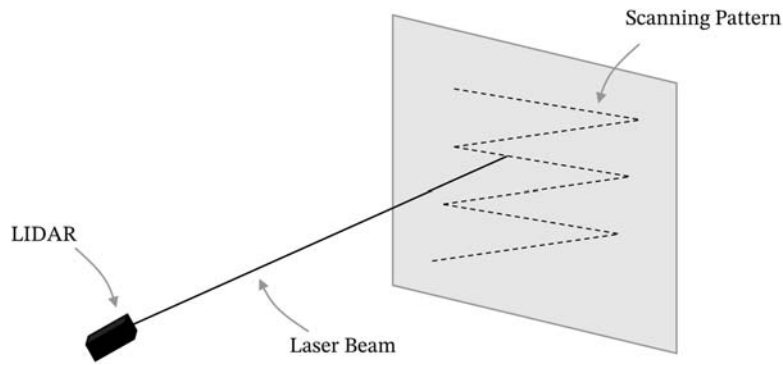


Figure 6.32 Scanning LIDAR.

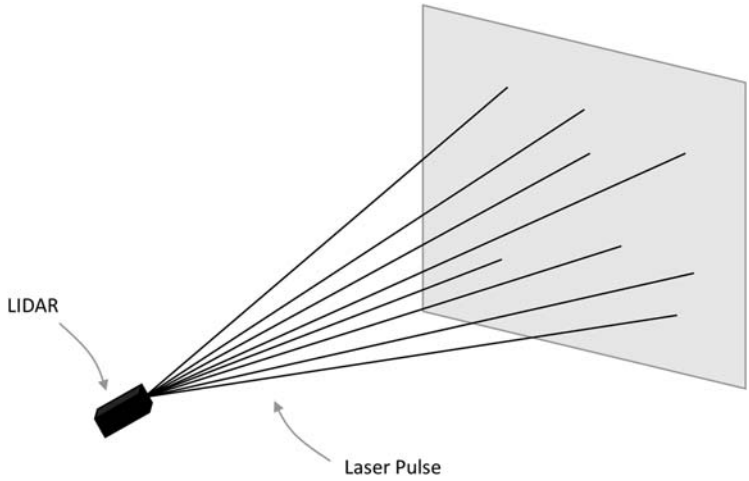


Figure 6.33 Flash LIDAR.

Table 6.15 LIDAR strengths and weaknesses.

Strengths	Weaknesses
Robust to operations with poor illumination conditions	High cost
Easy segmentation	High power consumption
3D position measurements	Complex hardware
	High computational cost

Altimeters

Satellite altimetry was developed in the early 60s, and it has generally been used to provide accurate measurements of the shape of a planetary surface. This kind of measurement can be widely applied to oceanography, geodesy, and geophysics. The altimeter working principle is to measure the travel time of an electromagnetic signal from the spacecraft to the planetary surface and back. This is common to all different kinds of altimeters, but the properties of the signal can infer its class: single beam, multibeam, or scanning synthetic aperture system. Furthermore, the operating frequency of the instrument implies a further classification, i.e., radar and laser. For science applications, scanning and multibeam systems are widely used. On the

contrary, in the GNC domain, mostly single beam altimeters are adopted. In this chapter, the main characteristics of single beam radar and laser altimeters are presented.

Altimetry principles

The objective of satellite altimetry is to measure the range between the satellite and the planet surface. A primary design parameter of an altimeter system is the area on the planet surface over which the range from the altimeter to the reference surface is measured. Intuitively, this can be approximated as the antenna footprint. The footprint of an antenna is traditionally defined, in a *beam-limited* sense, as the area on the target surface within the field of view subtended by the beamwidth of the antenna gain pattern. Thus, narrow beams have a higher accuracy, but they would require very large antennae that are not viable for space missions. Moreover, this kind of functional mode is more sensible to mispointing. In a *pulse-limited* configuration, the same results can be achieved with an antenna with a smaller diameter. This is possible because the return pulse is dictated by the length of the emitting pulse as shown in Fig. 6.34. In other words, the radiated phase front meets the scattering surface over an area that is limited by the width of the transmitted pulse.

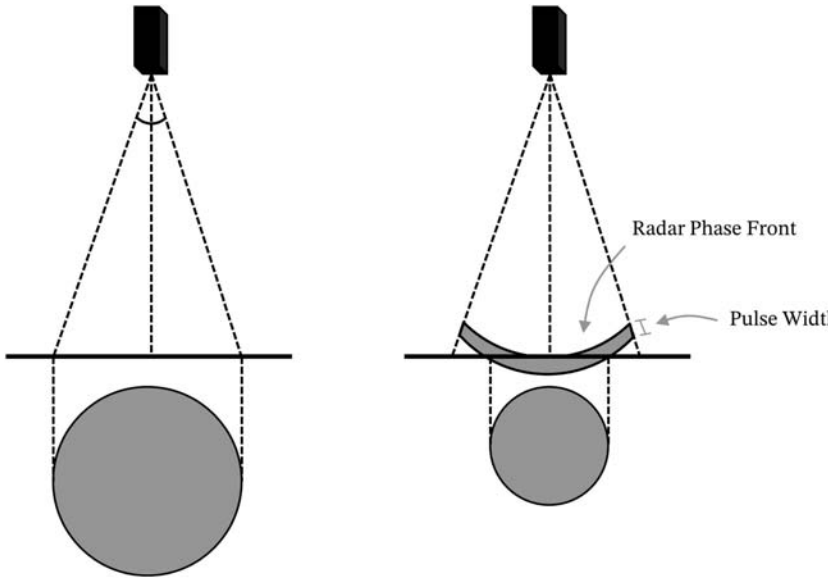


Figure 6.34 Beam-limited (left) and pulse-limited (right) footprint.

All the radar altimeters flown in space are pulse-limited, except for laser altimeters that are usually beam-limited due to their higher working frequency.

Radar and laser altimeters

The main difference between radar and laser altimeter is their working frequency. Both approaches have their strength and weakness that are summarized in Table 6.16.

As previously detailed, radar altimeters have a larger footprint with respect to laser altimeters. This implies a lower "horizontal" resolution but a comparable "vertical" accuracy. It is important to consider that radar altimeters do not suffer of attenuation in the presence of atmosphere. This is usually relevant for Earth applications (laser altimeters are weather dependents), but it may be neglected for some space applications. Size and mass are comparable, but radar antennae are usually larger and heavier but with a limited power consumption.

While several past Moon and Mars missions relied on radar altimeters [45,46], the current trend is to move toward laser-based sensors, such as LIDARs [47]. Despite the kind of altimeters, some aspects directly related to GNC can be highlighted. An advantage of using altimeters as additional sensors is that altitude measurements can be continuously collected without sunlight or visibility restrictions. However, an altimeter provides measurements relative to the surface, which may have an unknown height above the planet's center of mass; this is particularly true for small-body and unknown targets. In these cases, a shape model needs to be estimated or updated simultaneously with the spacecraft's orbit during the initial approach phase. Note that altimeters are extremely useful in GNC for

Table 6.16 Radar versus laser altimeter.

	Radar altimeter	**Laser altimeter**
Footprint	km level	m level
Vertical accuracy	Tens of centimeters	Tens of centimeters
Atmospheric Attenuation	Yes	No
Size and mass	Higher	Lower
Power consumption	Lower	Higher

landing applications, since in this case, the relative distance with respect to the surface is the fundamental quantity to measure.

Altimeter model

A simple model of a single beam altimeter can be implemented by considering a single ray hitting a spherical surface of a planet or an astral body. A simplified ray-tracing algorithm can be employed to find the intersection between the altimeter beam and the spherical body's surface. Finding intersection points between a ray and a sphere is a known problem, and it can be tackled by solving a quadratic equation:

$$\mathbf{A}t^2 + \mathbf{B}t + \mathbf{C} = 0,$$

with:

$$\mathbf{A} = \mathbf{d} \cdot \mathbf{d}$$

$$\mathbf{B} = 2\mathbf{p} \cdot \mathbf{d}$$

$$\mathbf{C} = \mathbf{p} \cdot \mathbf{p} - R^2$$

where **d** is the direction of the ray, **p** is the vector from the origin of the ray to the center of the sphere, t is the length of the ray, and R is the radius of the body. This quadratic equation has two solutions, corresponding to the two points on the sphere as in Fig. 6.35.

Furthermore, the determinant of the quadratic equation ($\Delta = B^2 - 4AC$) helps identifying three different cases:

- $\Delta < 0$: there is no intersection with the sphere.
- $\Delta = 0$: there is only one solution of the equation, i.e., the ray is tangent to the sphere surface.
- $\Delta > 0$: there are two solutions corresponding to the two segments $\overline{SP_1}$ and $\overline{SP_2}$ with respect to the sensor position S.

  The 3D points in the planet centered frame can be computed as:

$$\mathbf{P}_1 = \mathbf{p} + \mathbf{d}^* \overline{SP_1}$$

$$\mathbf{P}_2 = \mathbf{p} + \mathbf{d}^* \overline{SP_2}$$

In order to determine which point is actually observed, the projections of the two vectors onto the beam direction are compared and the smallest one is taken as observed point.

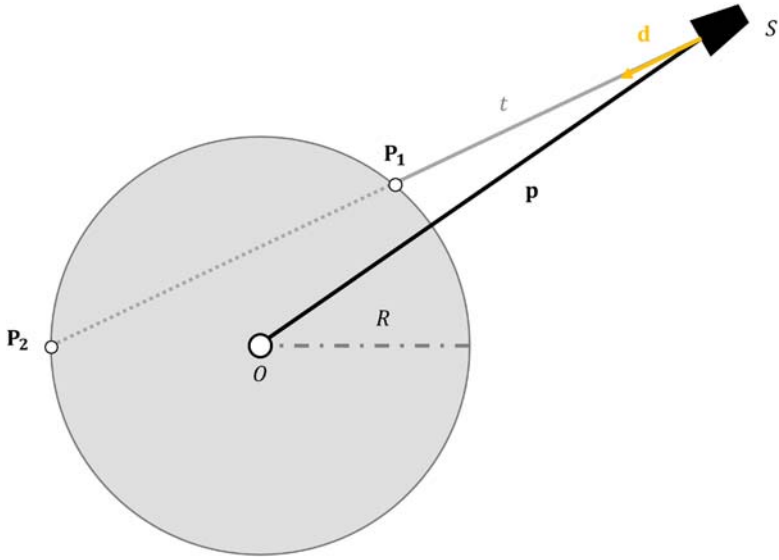


Figure 6.35 Ray-sphere intersection.

An alternative approach exploits a more complex model, considering the actual texture and roughness of the surface of the planet. This is done by performing a ray tracing with a 3D model of the observed body. This operation cannot be described by a simple mathematical model, and ad hoc solutions should be implemented.

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